

The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols

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Abstract

Horn's problem is concerned with characterizing the eigenvalues (a, b, c) of a Hermitian matrices (A, B, C) satisfying the constraint $A + B = C$ and has deep connections to tensor product invariants, Littlewood-Richardson coefficients, geometric invariant theory and the intersection theory of Schubert varieties. This paper concerns a generalization of Horn's problem we call the *tetrahedral Horn problem* that aims to characterize the tuples of eigenvalues (a, b, c, d, e, f) of Hermitian matrices (A, B, C, D, E, F) forming the edges of a tetrahedron in the space of Hermitian matrices and thus satisfying the constraints $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. We derive new inequalities satisfied by the Schur-polynomials of such eigenvalues and, using eigenvalue estimation techniques from quantum information theory, prove their satisfaction up to degree k implies the existence of approximate solutions with error $O(\sqrt{\frac{\log k}{k}})$. Moreover, the existence of such a matrices is connected to the semiclassical asymptotics of the $6j$ -symbols for the unitary group $U(n)$, which are maps between multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\delta \cong V_\alpha \otimes (V_\beta \otimes V_\delta)$ for unitary irreducible representations. Using our techniques, we prove the decay rate of the norms of such $6j$ -symbols is either inverse-polynomial or exponential depending on whether there exists such tetrahedra of Hermitian matrices.

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1 Introduction

Horn’s problem: The problem of determining the set of eigenvalues (a, b, c) of Hermitian matrices (A, B, C) satisfying $A + B = C$ has been investigated for over a century [Wey12; Fan49; Lid53; Wie55; TF71; TF70] and has come to be known as *Horn’s problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. In 1998, Klyachko provided a solution to Horn’s problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn’s original conjecture [Ful98]. These linear inequalities, now known as the *Horn inequalities* describe the convex polytope of admissible values for (a, b, c) . Recent developments on Horn’s problem include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues c of the sum $C = A + B$ if A and B are sampled uniformly and independently from the respective sets of Hermitian matrices with eigenvalues a and b by Coquereaux, McSwiggen, and Zuber in series of papers [CZ18; CZ18; CMZ19; Zub18]. For a pedagogical and historical introduction to Horn’s problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00] or alternatively [NP06, Appendix D]. See also the article by Daftuar and Hayden for an account of applications to topics in quantum information theory [DH05].

In parallel to Horn’s problem, variations and extensions of Horn’s problem have been explored by numerous authors. For example, Agnihotri and Woodward, Belkale, and [BK16] have studied the eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00; AMW01].

The tetrahedral Horn problem: In this paper, we consider a generalization of Horn’s problem investigated previously only briefly in [CŠW18]. While Horn’s problem aims to characterize the eigenvalues (a, b, c) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization we consider here aims to characterize the possible eigenvalues (a, b, c, d, e, f) of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints:

$$A + B = C, \quad B + D = F, \quad D + C = E, \quad \text{and} \quad A + F = E. \quad \text{\texttt{\{eq:tet_cond\}}} \quad (1)$$

These constraints arise naturally from building a tetrahedral arrangement of $n \times n$ Hermitian matrices inside the vector space of such Hermitian matrices with edges labelled according to Fig. 1. For this reason, we refer to this eigenvalue problem as the *tetrahedral* Horn problem and we say a sextuple of eigenvalues, (a, b, c, d, e, f) , is *tetrahedral* if such matrices exist, and not tetrahedral otherwise. Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper.

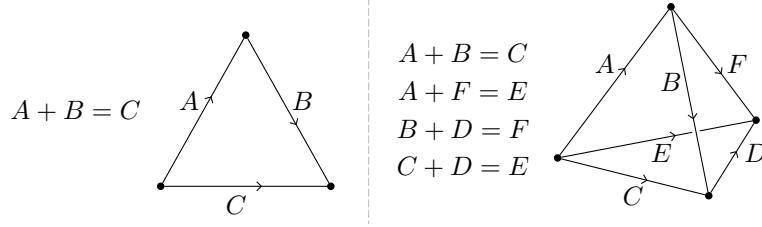


Figure 1: (Left) Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. fig:tetrahedron_labelling_scheme

It is well-known that the space of admissible eigenvalues for a triangular arrangement of 2×2 traceless Hermitian matrices is equivalent to the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. This correspondence generalizes to the tetrahedral case in the sense that the space of admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is equivalent to the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While these correspondences are straightforward to prove and also well-known, this geometric knowledge about the solution space leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices; specifically, the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is described by the conjunction of the triangle inequalities for each of the four faces (these inequalities coincide with the solution to Horn's original problem) together with the positivity of so-called Cayley-Menger determinants which are polynomial inequalities in the side-lengths (a, b, c, d, e, f) [SJS18]. Even for $n = 2$, we remark that there is a strong qualitative difference between the tetrahedral Horn problem and the original Horn problem. In particular, while the Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is not convex nor a polytope.

Our contributions: Regarding the tetrahedral Horn problem, we derive a new family of inequalities constraining the set of tetrahedral eigenvalues. These inequalities are expressed in terms of Schur polynomials $s_\lambda(x)$ of the involved eigenvalues which are homogeneous symmetric polynomials indexed by Young diagrams λ . More accurately, we find our inequalities are best articulated in terms of a simple rescaling of the Schur polynomials, $\varphi_\lambda(x) := s_\lambda(x)/H_\lambda$, where H_λ is a positive integer equal to the product of the hook-lengths of all boxes in the Young diagram λ . These inequalities are presented in the following theorem. thm:iff_tet

Theorem 1. *Let (a, b, c, d, e, f) be a sextuple of non-negative eigenvalues of length n satisfying the trace conditions:*

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \text{Tr}[b] + \text{Tr}[d] = \text{Tr}[f], \text{Tr}[c] + \text{Tr}[d] = \text{Tr}[e], \text{ and } \text{Tr}[a] + \text{Tr}[f] = \text{Tr}[e]. \quad (2)$$
{eq:trace_conditions}

Then there exists Hermitian matrices (A, B, C, D, E, F) with satisfying $A + B = C$, $B + D = F$, $C + D = E$ and $A + F = E$ with eigenvalues (a, b, c, d, e, f) if and only if for all triples of Young diagrams (α, β, δ) ,

$$\varphi_\alpha(a)^{\frac{2}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (3)$$
{eq:nec_tet}

The expression $\left\{ \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\}$ appearing above is an operator known as the $6j$ -symbol for the group of unitary $n \times n$ matrices $U(n)$ whose significance will be explained shortly. While Theorem 26 provides a complete characterization of the set of tetrahedral eigenvalues, it does so using an infinite number of inequalities. Nevertheless, we prove the finite subset of such inequalities of fixed degree $k = |\alpha| + |\beta| + |\delta|$ is sufficient to characterize the set of sextuples of eigenvalues, (a, b, c, d, e, f) , which are *approximately* tetrahedral in the sense that their distance, $D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right)$, to the set of tetrahedral eigenvalues is small.

Theorem 2. If (a, b, c, d, e, f) are non-negative eigenvalues satisfying the trace conditions in Eq. (2) and also the inequality in the inequality in Eq. (105) for all triples (α, β, δ) such that $|\alpha| + |\beta| + |\delta| = k$ for some positive integer k , then

$$\frac{1}{\text{Tr}[e]} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq O \left(n \sqrt{\frac{\log k}{k}} \right), \quad \text{\texttt{eq:tri_abd_cef_dist_bound}} \quad (4)$$

where $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ is the distance measure

$$D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \min_{(A,B,D) \sim \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \|\text{eig}(A+B) - c\|_1^2 + \|\text{eig}(B+D) - f\|_1^2 + \|\text{eig}(A+B+D) - e\|_1^2. \quad \text{\texttt{eq:defn_tet_deviation}} \quad (5)$$

Note that $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} = 0$ if and only if (a, b, c, d, e, f) are tetrahedral.

Note this theorem actually implies the “if” portion of Theorem 26 as one can take $k \rightarrow \infty$. Along the way, we derive an analogous set of inequalities for the original Horn problem along with an accompanying distance bound (see Proposition 59).

6j-symbols: Given a complex reductive group G with unitary irreducible representations (irreps) V_λ labelled by highest weights λ along with tensor-product multiplicity spaces, $M_{\alpha,\beta}^\gamma := \text{Hom}_G(V_\gamma, V_\alpha \otimes V_\beta)$, the 6j-symbols are maps between these multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\gamma \cong V_\alpha \otimes (V_\beta \otimes V_\gamma)$ between a triple of irreps, V_α, V_β , and V_δ of G . Specifically, a 6j-symbol depends on six irreps, $V_\alpha, V_\beta, V_\gamma, V_\delta, V_\epsilon$, and V_ϕ , and is the linear map between multiplicity spaces of the type

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon \rightarrow M_{\alpha,\phi}^\epsilon \otimes M_{\beta,\delta}^\phi,$$

corresponding to the block-wise components of the canonical isomorphism from $\bigotimes_\gamma M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$ to $\bigotimes_\phi M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$. Whenever the group G is multiplicity-free, in the sense that the dimensions of all multiplicity spaces, $m_{\alpha,\beta}^\gamma := \dim M_{\alpha,\beta}^\gamma$, are either zero or one, the corresponding 6j-symbol is merely a scalar quantity.

Wigner’s 6j-symbol for SU(2): The group SU(2) is multiplicity-free and the 6j-symbols for SU(2) are, depending on the normalization and/or phase convention, are also referred to as Wigner 6j-symbols or Wigner 6j-coefficients [Wig93] or the Racah W -coefficients [Rac42]. The 6j-symbols for SU(2) play a foundational role in the quantum theory of angular momentum, modelling the recoupling theory of three angular momenta, and thus they are used frequently in calculations involving the scattering of multiple particles in nuclear, atomic and molecular physics [VMK88; Edm96; Aqu+12]. Beyond the direct application to the theory of angular momenta, the 6j-symbols for SU(2) also constitute a primitive object in the theory of spin networks [Pen71; BS10], representing a minimal non-trivial closed spin network. General spin networks are networks of trivalent vertices and connections labelled by irreps (spins) of SU(2) which provide a discrete/combinatorial and manifestly invariant model with applications to gauge theories on graphs or manifolds [Bae96], invariants of knots and manifolds [Res87; TV92], topological quantum field theory [DJN93], elementary gates in spin network models for quantum computing [MR05], quantum change-point discrimination [AH11], and various aspects of proposed models for quantum gravity [Bae99; BC98; Bar03; Rov04; Mou84].

Throughout this paper, by the *asymptotics* of 6j-symbols, we are referring to their behaviour in the regime wherein the six highest weights, $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$, are becoming larger and larger in a suitable sense. The asymptotics of the 6j-symbol for SU(2), wherein the irreps of dimension $\ell + 1$ may be identified either by integers $\ell \in \{0, 1, \dots\}$ or half-integers $j = \frac{\ell}{2} \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, were initially investigated by Wigner [Wig60, Chapter 27]. In the context of the quantum theory of angular momenta, the limit of large ℓ , or equivalently large spins j , may also be interpreted as the limit wherein Planck’s constant tends to zero, $\hbar \propto \frac{1}{k} \rightarrow 0$, which suppresses the effects of quantizing angular momentum; therefore, these asymptotic limits are also known as *semiclassical* limits. Given three particles with spins $\frac{a}{2}$, $\frac{b}{2}$ and $\frac{d}{2}$ and total spin $\frac{e}{2}$, the 6j-symbol, $\left\{ \begin{matrix} a & b & c \\ d & e & f \end{matrix} \right\}$, can be interpreted as the quantum mechanical amplitude to observe the first and second particles to have combined spin $\frac{c}{2}$ given the second and third have combined spin $\frac{f}{2}$ [Wig60; Rob99]. Wigner

considered the probability for this event to occur in the asymptotic limit of large spins, i.e., the square of the $6j$ -symbol, and conjectured the estimate:

$$\left\{ \begin{matrix} a & b & c \\ d & e & f \end{matrix} \right\}^2 \approx \frac{1}{3\pi V}$$

where V is the volume of the Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) . Building upon the work of Wigner, Ponzano and Regge conjectured a formula for the amplitude of the $6j$ -symbol itself in the limit of large spins [PR68]. Later, using techniques from geometric quantization theory, Roberts rigorously proved an asymptotic formula which is equivalent to the conjecture of Ponzano and Regge (up to minor differences) [Rob99].

Theorem 3 (Roberts). *If (a, b, c, d, e, f) are integer side-lengths of a Euclidean tetrahedron with volume V and corresponding exterior dihedral angles¹ $(\theta_a, \theta_b, \theta_c, \theta_d, \theta_e, \theta_f)$, then*

$$\left\{ \begin{matrix} ka & kb & kc \\ kd & ke & kf \end{matrix} \right\} \stackrel{k \rightarrow \infty}{\sim} \sqrt{\frac{2}{3\pi V k^3}} \cos \left(\sum_{\ell} (k\ell + 1) \frac{\theta_{\ell}}{2} + \frac{\pi}{4} \right),$$

where the sum is over the six sides ℓ . Otherwise, if there does not exist a Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) , then $\left\{ \begin{matrix} ka & kb & kc \\ kd & ke & kf \end{matrix} \right\}$ decays to zero at an exponential rate as $k \rightarrow \infty$.

A proof of the Ponzano-Regge asymptotic formula was also obtained by Chen, Ismail, and Simeonov [CIS99] (with a correction by Watson [Wat99]) using estimates from the theory of hypergeometric functions. Additional proofs of this asymptotic correspondence between $6j$ -symbols for $SU(2)$ and Euclidean tetrahedra have emerged in recent years including the elementary proof by Gurau which applies Stirling's approximation directly to Racah's single-sum formula [Gur08] as well as group integral approach to evaluate so-called simple spin networks as Feynman graphs [FK00; FL03]. See also the paper by Bartlett and Nzaganya for an accessible presentation of these kinds of calculations [BN23]. Analogous asymptotic formulas have also been obtained by Taylor and Woodward for the $6j$ -symbol associated to the quantum group $U_q(\mathfrak{sl}_2)$ for a particular k -dependent parameter q [TW06]. In this generalized setting, the asymptotics calculated by Taylor and Woodward encode (i) the existence of a spherical tetrahedron on the unit sphere with prescribed edge-lengths whenever $q = \exp(i\pi/r(k))$ is an $r(k)$ th root of unity or (ii) the existence of a hyperbolic tetrahedron with prescribed edge-lengths when $q = \exp(\pi/r(k))$ is positive real (here $r(k) = k(r-2) + 2$ for $r > 2$ a fixed integer) [TW06].

More generally, that the asymptotics of representation-theoretic quantities (such as $6j$ -symbols) are controlled by the existence of some critical geometric object is explained by the theory of geometric quantization [Rob02]. Geometric quantization, broadly speaking, associates classical phase spaces modelled by symplectic manifolds to quantum phase spaces modelled by Hilbert spaces [Rob02; GS82; Kir04]. In particular, Kirillov's orbit principle [Kir04] proposes an association between (i) the coadjoint orbits of a compact Lie group G passing through an integral weight λ to (ii) the corresponding unitary irreps of G of highest weight λ . Furthermore, products of coadjoint orbits correspond to tensor products of irreps and moreover symplectic quotients of the resulting products of orbits are identified with invariant vectors in the resulting tensor products of irreps [Rob02; GS82]. For the aforementioned case of $SU(2)$, the coadjoint orbits can be identified the $SO(3)$ rotations of vectors in Euclidean $\mathbb{R}^3$ (i.e., spheres of fixed radius), and thus the asymptotics of representation-theoretic quantities for $SU(2)$ are controlled by the existence of corresponding geometric objects in Euclidean $\mathbb{R}^3$ with prescribed edge-lengths [Rob02].

$6j$ -symbols for $U(n)$: One application of the techniques developed in this paper is to provide estimates for the asymptotics of the $6j$ symbols for the group $U(n)$ (or equivalently for $SU(n)$) wherein the polynomial irreps V_{λ}^n are labelled by highest-weights which can also be identified with Young diagrams λ . One of the primary difficulties in proving statements about $6j$ symbols for $U(n)$ for $n \geq 3$ is the presence of non-trivial multiplicity spaces, meaning the dimensions of tensor-product multiplicity spaces for $U(n)$, also known as Littlewood-Richardson coefficients, can be greater than one. Because of these difficulties, over the years,

¹For example, the exterior dihedral angle θ_a is the outside angle between the two faces of the tetrahedron sharing the common edge with side-length a .

considerable attention has been placed on the study of $U(n)$ (or $U_q(\mathfrak{sl}_n)$) $6j$ symbols which are multiplicity-free in the sense that the dimensions of each of the multiplicity spaces is at most one [NPZ13; AMS20; Alc+23]. Indeed, as stated by Coquereaux, “apart from very few special cases (low dimension of irreps and no multiplicity), ... we are not aware of any published general result.” [Coc23].

Our approach bypasses the difficulties presented by multiplicities by focusing only on the *norms* of the $6j$ symbol for $U(n)$. Following the procedure provided by geometric quantization, one expects the asymptotics of the $6j$ -symbol for $U(n)$ to be controlled by existence of some kind of tetrahedral object in space of coadjoint orbits for $U(n)$. In particular, the coadjoint orbits of $U(n)$ can be identified with the orbits of $n \times n$ Hermitian matrices, X , under the conjugation action, $X \mapsto U^{-1}XU$, by unitary $n \times n$ matrices $U \in U(n)$; in other words, coadjoint orbits are associated to sets of $n \times n$ Hermitian matrices sharing a common set of eigenvalues $x = (x_1 \geq \dots \geq x_n)$. Using our methods we are able to prove the following inverse polynomial lower bound. cor: asym_6j

Corollary 4. *If (a, b, c, d, e, f) are the eigenvalues of $n \times n$ of positive semidefinite matrices (A, B, C, D, E, F) satisfying $A + B = C$, $B + D = F$, $C + D = E$, and $A + F = E$, then there exists a sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ (with $|\epsilon_k| = k$) converging in proportion to the eigenvalues (a, b, c, d, e, f) such that*

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\| \geq \frac{1}{\text{poly}(k)}. \quad (6)$$

Otherwise, if (a, b, c, d, e, f) are not the eigenvalues of any such matrices, then for any sequence of Young diagrams converging in proportion to (a, b, c, d, e, f) , we have exponential decay in the value of $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|$.

Before continuing, it is worth emphasizing the similarities and differences between our result and the asymptotic formula proven by Roberts above [Rob99]. First and foremost, our result goes beyond that of Roberts in the sense that it applies to the higher-rank Lie group $U(n)$ for $n \geq 3$. Moreover, our result, even when restricted to the case for $SU(2)$, goes beyond the statement proven by Roberts in the sense our result is applicable to *irrational* eigenvalues/edge-lengths, whereas the former is only applicable to *integer*, or with a minor re-scaling, *rational* eigenvalues/edge-lengths. Like the Pozzano-Regge formula, our result is different in that it merely promises the *existence* of a sextuple of sequences of Young diagrams, $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, converging to the eigenvalues (a, b, c, d, e, f) , whereas the result of Roberts provides a concrete statement about the particular sequence of *scaled* Young diagrams $(k\alpha, k\beta, k\gamma, k\delta, k\epsilon, k\phi)$. Moreover, our result provides a polynomial lower bound on the value of the $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ and does not provide an asymptotic formula which would serve as the proper generalization of the Ponzano and Regge formula above [PR68]. The Ponzano and Regge formula for $SU(2)$ was traditionally obtained using the method of *stationary phases* which relies on both an integral formula for the $6j$ -symbol with exponential integrand along with a geometric understanding of the *isolated* critical points for the exponential phase [PR68; Gur08]. In the case of $SU(2)$ with generic edge-lengths, there is a formulation involving exactly *two* isolated critical points corresponding the $SO(3)$ -orbit $[\tau]$ of a Euclidean tetrahedron τ with given edge-lengths and (ii) the $SO(3)$ -orbit $[\tau']$ of the Euclidean tetrahedron τ' which is the mirror-image of the tetrahedron τ with the same fixed side lengths. Generalizing this type of criticality analysis to $U(n)$ for $n \geq 3$ appears to be a significant challenge, and even if an analysis was performed, one would still need to compute a corresponding Hessian for each critical point and evaluate the total contribution [Rob02]. Perhaps a different approach for deriving the Ponzano-Regge formula, such as those based on recursion relations or estimates of hypergeometric functions appearing in [CIS99; TW06; Art23; Art24] would better suited for the case where $n \geq 3$.

Eigenvalue estimation: Our approach to proving the above theorems relies on the technique of *eigenvalue estimation*, a probabilistic technique developed in the context of quantum information theory for estimating the eigenvalues of positive semidefinite matrices which model the state of quantum systems [KW01; CM06; OW16; HM02; CHW07; Hay17]. The eigenvalue estimation technique relies heavily on the concept of Schur-Weyl duality which describes the decomposition of the k -th tensor power of the n -dimensional complex vector space $\mathbb{C}^n$, which supports representations of both the symmetric group S_k by permutation of tensor factors and the group of unitary matrices $U(n)$ (or more generally invertible complex matrices $GL(n, \mathbb{C})$), into irreducible invariant subspaces indexed by Young diagrams $\lambda = (\lambda_1, \dots, \lambda_n)$ with k boxes and at most n rows. Given a positive semidefinite matrix X with eigenvalues $x = (x_1, \dots, x_n)$, one can analogously decompose the k -th tensor power of X into the same irreducible invariant subspaces indexed

by λ and then investigate the aptly named Schur-Weyl distribution, $p_x(\lambda) := \text{Tr}[P_\lambda X^{\otimes k}] / \text{Tr}[X^{\otimes k}]$, which measures the relative proportion of the decomposed pieces of $X^{\otimes k}$ and depends only on the eigenvalues x of X . The technique of eigenvalue estimation relies on the fact that the Schur-Weyl distribution concentrates strongly around those Young diagrams λ which are proportional to eigenvalues x and that this concentration strengthens as the sizes of the Young diagrams grow to infinity [KW01; CM06; OW16]. Applying the eigenvalue estimation technique to the original Horn problem was first done by Christandl [Chr08]. In this paper, we develop a technique called *eigenvalue separation* which relates the similarity between two Schur-Weyl distributions, $p_x(\lambda)$ and $p_y(\lambda)$, to the similarity of the underlying eigenvalues, x and y , and apply this technique to tetrahedral Horn problem and obtain Theorem 26. In addition, our approach produces new results for the original Horn problem, analogous to Theorem 26, beyond those obtained in [Chr08].

Quantum marginals & the $6j$ -symbol for the symmetric group S_k : Our method for proving Corollary 4 is closely related to techniques used in an earlier paper by Christandl, Şahinoğlu, and Walter which derived analogous bounds on the $6j$ symbol for the symmetric group S_k [CŞW18]. In that paper, the authors considered the problem of determining whether there exists a density matrix ρ_{ABC} with partial traces $\rho_{AB}, \rho_{BC}, \rho_A, \rho_B$, and ρ_C with a given sextuple of eigenvalues [CŞW18]. Such problems arise naturally in the context of quantum information theory and are typically referred to as quantum marginals problems [CŞW18; Fra22]. In particular, they proved the $6j$ symbols for the symmetric group S_k (referred to as the recoupling coefficients) in the limit as $k \rightarrow \infty$ either decay exponentially or can be bounded from below by a polynomial factor depending on whether there exists a density matrix with the prescribed marginal spectra [CŞW18]. Moreover, they also demonstrated how the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem and, in a complementary manner, furthermore demonstrated how, by restricting to the subgroup $S_k \subset U(k)$ of permutation matrices and applying Nishiyama’s restriction theorem [Nis00], the $6j$ symbols for unitary group $U(n)$ may be viewed as a special case of the $6j$ symbols for the symmetric group S_k [CŞW18]. This latter result obtained by [CŞW18] is a generalizes how the Littlewood-Richardson coefficients can be cast as a special case of the Kronecker coefficients [Lit58; Mur55; IP23]. While the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem, the approach taken by this paper differs from [CŞW18] as it tackles the tetrahedral Horn problem directly, and consequently, we obtain Theorem 26 which provides stronger conditions for tetrahedral eigenvalues which go beyond the conditions previously obtained in [CŞW18]. We note that this direct approach was initiated, for the $n = 2$ case, by Backens [Bac10].

Multiple Horn problems: More recently, Alekseev et al. have investigated a few variants of the tetrahedral Horn problem and are referred to as *multiple* Horn problems in [Ale+25]. Specifically, Alekseev et al. focus on two variants, the *multiplicative multiple Horn problem* concerning the singular values of products of invertible matrices, and the *tropical multiple Horn problem* concerning singular values of concatenations of weighted planar networks [Ale+25]. The tetrahedral Horn problem we consider here is termed the *additive* multiple Horn problem in the language of [Ale+25].

Other inequalities: It is worth highlighting that there exists related results constraining the eigenvalues of sums of Hermitian matrices which are already readily applicable to the tetrahedral Horn problem. In addition to the aforementioned Horn’s inequalities [Hor62], which constraint the eigenvalues of any triple of Hermitian matrices $(A, B, A + B)$, there exists a myriad of inequalities which constrain the eigenvalues of the quadruple of Hermitian matrices $(Y, X + Y, Y + Z, X + Y + Z)$ whenever $X, Y, Z \geq 0$ are positive semidefinite matrices (see for example [BS15; TCL11; Sag13; BGS08; FG13; LS14; Sra]). These inequalities are readily applicable to the tetrahedral Horn problem under the identification ($X = A, Y = B$ and $Z = D$) wherein they constrain, respectively, the eigenvalues (b, c, e, f) of the matrices (B, C, E, F) while ignoring the eigenvalues of A and D . More scarce are inequalities which additionally incorporate the eigenvalues (a, d) of the matrices (A, D) such as Minkowski-like determinant inequality [NS23, Section 6.2] (originally due to Courtade[Cou16; htt]), or applications of Ptolemy’s inequality [Sch52] or the Cayley-Menger determinant inequality [SJS18, Equation 5.2] to the space of $n \times n$ Hermitian matrices equipped with the Hilbert-Schmit inner product. While each of these previously known inequalities can be viewed as necessary conditions for the tetrahedral Horn problem, to the best of our knowledge, they remain insufficient for the tetrahedral Horn problem when $n \geq 3$. Our contribution, therefore, is to provide a family of necessary and sufficient

inequalities in the form of Theorem 26 which is applicable to all dimensions n .

Organization of the paper: In Section 2 we formally define the tetrahedral Horn problem, describe some of its symmetries, and provide a solution for the case of 2×2 matrices.

Anton's: TODO:

TC's: Then, in Section 5, we develop the mathematical machinery needed to prove the *necessity* of the inequalities in Theorem 26, specifically the binomial theorem for representations (Theorem 33). Afterwards, Section 6 is dedicated to proving the distance bound in Theorem 2 and thus the *sufficiency* of the inequalities in Theorem 26. The results in Section 6 rely on the results of Section 5, the aforementioned eigenvalue concentration bounds (Section 6.1), and some standard results from random matrix theory (Section 6.2).

2 The tetrahedral Horn problem

sec:tet_horn_prob

2.1 Eigenvalue problems for matrix sums

Throughout this paper, we consider the relationship between $n \times n$ Hermitian matrices and their real-valued tuples of eigenvalues. To denote the set of $n \times n$ Hermitian matrices, we use

$$\mathcal{H}_n := \{X \in M_n(\mathbb{C}) \mid X^\dagger = X\}, \quad (7)$$

and to denote the set of possible eigenvalues of $n \times n$ matrices, we use

$$\text{Eig}(n) := \{x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1 \geq \dots \geq x_n\}. \quad (8)$$

Let $\mathcal{H}_n^+$ be the subset of positive semidefinite $n \times n$ Hermitian matrices and $\text{Eig}^+(n)$ the corresponding set of non-negative eigenvalues. Given a Hermitian matrix $X \in \mathcal{H}_n$, $\text{eig}(X)$ denotes the eigenvalues of X . The set of Hermitian matrices with eigenvalues $x \in \text{Eig}(n)$ is denoted by

$$\mathcal{O}_x := \{X \in \mathcal{H}_n \mid \text{eig}(X) = x\}, \quad (9)$$

and $[x]$ denotes the diagonal matrix with entries $(x_1, \dots, x_n)$ along the diagonal:

$$[x] := \text{diag}(x_1, \dots, x_n). \quad (10)$$

The set of Hermitian matrices with eigenvalues x can then be identified with the unitary orbit of the diagonal matrix $[x]$, i.e., $\mathcal{O}_x = \{U[x]U^{-1} \mid U \in \text{U}(n)\}$. Moreover, for any function of Hermitian matrices, f , which unitarily invariant in the sense that $f(X) = f(UXU^{-1})$ holds for all $X \in \mathcal{H}_n$ and $U \in \text{U}(n)$, we let $f[x]$ be shorthand for $f([x])$. For example,

$$\text{Tr}[x] := \text{Tr}([x]), \quad \text{and} \quad \text{rank}[x] := \text{rank}([x]). \quad (11)$$

prob:tri_problem

Definition 5 (Trigonal Horn problem). A triple $(a, b, c) \in \text{Eig}(n)^3$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple $(A, B, C) \in \mathcal{H}_n^3$ of $n \times n$ Hermitian matrices satisfying (i) the linear constraint

$$A + B = C, \quad (12)$$

{eq:tri_linear_constraints}

and (ii) the three eigenvalue constraints

$$\text{eig}(A) = a, \quad \text{eig}(B) = b, \quad \text{eig}(C) = c. \quad (13)$$

{eq:tri_eigenvalue_constraints}

The problem of determining the set of trigonal eigenvalues (a, b, c) is known as *Horn's problem* or the *original Horn problem*.

prob:tet_problem

Definition 6 (Tetrahedral Horn problem). A sextuple $(a, b, c, d, e, f) \in \text{Eig}(n)^6$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple $(A, B, C, D, E, F) \in \mathcal{H}_n^6$ of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad (14)$$

{eq:tet_linear_constraints}

and (ii) the six eigenvalue constraints

$$\text{eig}(A) = a, \quad \text{eig}(B) = b, \quad \text{eig}(C) = c, \quad \text{eig}(D) = d, \quad \text{eig}(E) = e, \quad \text{eig}(F) = f. \quad (\text{eq:tet_eigenvalue_constraints}) \quad (15)$$

The problem of determining the set of tetrahedral eigenvalues (a, b, c, d, e, f) will be referred to as the *tetrahedral Horn problem*.

The tetrahedral Horn problem can be viewed as a generalization of the Horn's problem in a variety of ways. For instance, a triplet (a, b, c) of eigenvalues is trigonal if and only if the sextuple $(a, b, c, 0, c, b)$ is tetrahedral. More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of the original Horn original problem subject to additional matching constraints. Specifically, a sextuple of eigenvalues (a, b, c, d, e, f) is tetrahedral only if each of the following four triplets are trigonal:

$$(a, b, c), \quad (b, d, f), \quad (c, d, e), \quad (a, f, e). \quad (\text{eq:four_faces}) \quad (16)$$

If each of the above triplets are trigonal, then there exists a set of twelve Hermitian matrices with the corresponding eigenvalues satisfying

$$A + B = C, \quad \tilde{B} + D = F, \quad \tilde{C} + \tilde{D} = E, \quad \tilde{A} + \tilde{F} = \tilde{E}. \quad (\text{eq:four_faces_matrices}) \quad (17)$$

The tetrahedral Horn problem goes beyond four separate instances of Horn's original problem in that it additionally requires the corresponding pairs of primed and unprimed Hermitian matrices above must coincide.

2.2 Low dimensional solutions

In the original Horn problem, the eigenvalues (a, b, c) of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \quad (\text{eq:tri_trace_cond}) \quad (18)$$

where $\text{Tr}[x] := \sum_i x_i$ is shorthand for the sum of the eigenvalues $x = (x_1 \geq \dots \geq x_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = \text{Tr}[x]$ for any Hermitian matrix X with eigenvalues x . For the tetrahedral Horn problem, a set of tetrahedral eigenvalues (a, b, c, d, e, f) necessarily satisfy four *trace conditions*:

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \quad \text{Tr}[b] + \text{Tr}[d] = \text{Tr}[f], \quad \text{Tr}[d] + \text{Tr}[c] = \text{Tr}[e], \quad \text{Tr}[a] + \text{Tr}[f] = \text{Tr}[e]. \quad (\text{eq:tet_trace_conds}) \quad (19)$$

Furthermore, for the trivial case where $n = 1$, the above trace constraints are both necessary and sufficient for the tetrahedral Horn problem.

exam:n_2

Example 7 (Tetrahedral Horn problem for $n = 2$). Here we articulate the solution to the tetrahedral Horn problem for 2×2 Hermitian matrices. We restrict our focus to the case of *traceless* 2×2 Hermitian matrices². In this context, the sextuple of eigenvalues (a, b, c, d, e, f) can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that the sextuple of traceless eigenvalues are

$$(a, -a), \quad (b, -b), \quad (c, -c), \quad (d, -d), \quad (e, -e), \quad (f, -f). \quad (\text{eq:traceless_two_dim_tetra}) \quad (20)$$

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (21)$$

Under this identification, the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|_2, -\|x\|_2), \quad \text{where,} \quad \|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (22)$$

²In Section 2.3, Corollary 13, we will see that this restriction can be made without loss of generality.

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently, in this setting, the tetrahedral Horn problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (\text{eq:quadruplet_triangle_constraints}) \quad (23)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$\text{CM}(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & a^2 & c^2 & e^2 & 1 \\ a^2 & 0 & b^2 & f^2 & 1 \\ c^2 & b^2 & 0 & d^2 & 1 \\ e^2 & f^2 & d^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \geq 0. \quad (\text{eq:volume_positive_constraint}) \quad (24)$$

The quantity $\text{CM}(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $\text{CM}(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $\text{CM}(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]³. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (23) and Eq. (24).

2.3 Symmetries of tetrahedral eigenvalues

sec:symmetries

There are numerous types of symmetries present in the statement of Definition 6. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Definition 6 as stated.

prop:scaling

Proposition 8 (Scaling). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, (sa, sb, sc, sd, se, sf) , are tetrahedral if and only if the unscaled eigenvalues (a, b, c, d, e, f) are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (14) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) , e.g., we have $sA + sB - sC = 0$ if and only if $A + B - C = 0$ because $s \neq 0$. $\square$

One may wonder whether it is possible to extend the scaling symmetry to negative values for s . Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrix both negates and reverses the sorting order of its eigenvalues.

prop:inversion

Proposition 9 (Inversion). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $x = (x_1 \geq \dots \geq x_n)$, let x^* denote the inverted list of resorted and negated eigenvalues: $x^* = (-x_n \geq \dots \geq -x_1)$. Then (a, b, c, d, e, f) are tetrahedral if and only if $(a^*, b^*, c^*, d^*, e^*, f^*)$ are tetrahedral.*

Proof. The statement holds because $\text{eig}(-X) = x^*$ whenever $\text{eig}(X) = x$ and moreover the sextuple of matrices (A, B, C, D, E, F) satisfies Eq. (14) if and only if the sextuple of negated matrices $(-A, -B, -C, -D, -E, -F)$ satisfies Eq. (14). $\square$

The inversion symmetry Proposition 9 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $x \rightarrow x^*$, or permutations of elements in the sextuple (a, b, c, d, e, f) .

³The triangle inequalities $|a - b| \leq c \leq a + b$ are also equivalent to the non-negativity of the *area*, $\text{CM}(a, b, c)$, of the triangle with side-lengths a, b and c where $\text{CM}(a, b, c)^2 = -\frac{1}{16} \det \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & a^2 \\ 1 & a^2 & 0 \end{pmatrix}$ is equivalent to Heron's formula.

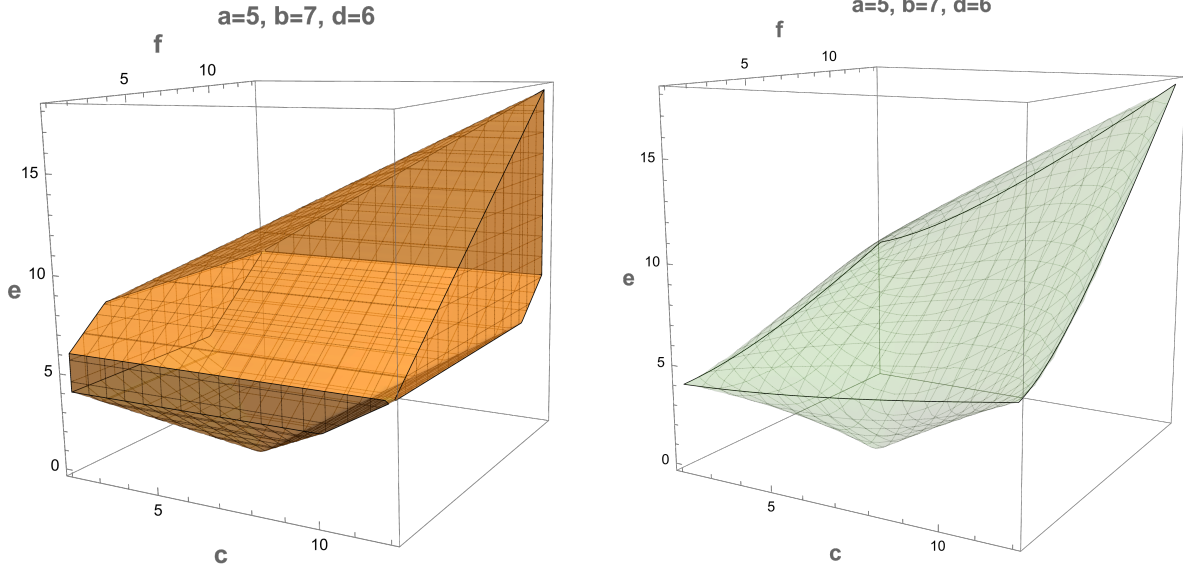


Figure 2: A comparison between the values for (c, e, f) with fixed $(a, b, d) = (5, 7, 6)$ which (left) merely satisfy the triangle inequalities applied to each face the tetrahedron (Eq. (23)) and (right) those which additionally satisfy the Cayley-Menger determinant inequality (Eq. (24)). The figure on the right depicts (a slice of) the moduli space of Euclidean tetrahedra with side lengths (a, b, c, d, e, f) and thus depicts the solution to the tetrahedral Horn problem for 2×2 traceless Hermitian matrices; remark that it is neither a polytope or convex. The figure on the left depicts (a slice of) a polytope, formed by the triangle inequalities (Eq. (23)) which forms a loose outer approximation to the figure on the right.

prop:tet_symmetry

Proposition 10 (Oriented). *The sextuple (a, b, c, d, e, f) of eigenvalues are tetrahedral if and only if the following sextuples of eigenvalues are tetrahedral:*

$$(a \ f \ e \ d^* \ c \ b) \quad (c \ e^* \ d^* \ f \ b \ a^*) \quad (25)$$

{eq:gens_of_tet_sym}

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{pmatrix} a & b & c & d & e & f \\ d^* & e & c & a^* & b & f \\ a & e^* & f^* & d & b^* & c^* \end{pmatrix} \begin{pmatrix} a^* & c & b & d & f & e \\ f^* & b & d^* & c^* & e^* & a^* \\ c & b^* & a & f & e & d \\ a & f & e & d^* & c & b \end{pmatrix} \begin{pmatrix} a & c^* & b^* & d^* & f^* & e^* \\ b & a & c & e^* & d^* & f^* \\ f & b^* & d & c & e & a \\ e^* & d & c^* & b & a^* & f \\ c^* & b & a^* & f^* & e^* & d^* \\ a^* & f^* & e^* & d & c^* & b^* \end{pmatrix} \begin{pmatrix} a^* & b^* & c^* & d^* & e^* & f^* \\ d^* & b^* & f^* & a^* & e^* & c^* \\ d & e^* & c^* & a & b^* & f^* \\ a^* & e & f & d^* & b & c \end{pmatrix} \begin{pmatrix} b & f^* & d^* & e & c & a \\ c & a^* & b & f^* & d^* & e^* \\ e & c^* & d & b & f & a^* \\ f & a & e & c^* & d & b^* \\ f^* & d & b^* & c & a & e \\ e^* & f & a^* & b^* & c^* & d \\ c^* & d^* & e^* & f & a^* & b \\ b^* & c & a & e^* & f^* & d^* \end{pmatrix} \begin{pmatrix} b & d & f & e^* & a^* & c^* \\ d & f^* & b^* & a^* & c^* & e^* \\ f & e^* & a^* & c & b & d^* \\ e^* & a & f^* & b & d^* & c \\ d^* & c^* & e^* & a & f^* & b^* \\ c^* & e & d & f^* & b^* & a \end{pmatrix} \begin{pmatrix} b^* & d^* & f^* & e & a & c \\ d^* & f & b & a & c & e \\ f^* & e & a & c^* & b^* & d \\ e & a^* & f & b^* & d & c^* \\ d & c & e & a^* & f & b \\ c & e^* & d^* & f & b & a^* \end{pmatrix} \begin{pmatrix} b^* & f & d & e^* & c^* & a^* \\ c^* & a & b^* & f & d & e \\ e^* & c & d^* & b^* & f^* & a \\ f^* & a^* & e^* & c & d^* & b \\ f & d^* & b & c^* & a^* & e^* \\ e & f^* & a & b & c & d^* \\ c & d & e & f^* & a & b^* \\ b & c^* & a^* & e & f & d \end{pmatrix} \quad (26)$$

{eq:all_of_tet_sym}

Proof. To begin, notice that each of the operations on (a, b, c, d, e, f) appearing in Eq. (25) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. Now suppose (a, b, c, d, e, f) are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (14). Then, for example, the sextuple $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the

eigenvalues (a, f, e, d^*, c, b) is also tetrahedral because

$$\begin{aligned} A' + B' - C' &= (A + F - E) = 0, \\ B' + D' - F' &= -(B + D - F) = 0, \\ C' + D' - E' &= -(C + D - E) = 0, \\ A' + F' - E' &= (A + B - C) = 0. \end{aligned} \tag{27}$$

Moreover, as the operation mapping (a, b, c, d, e, f) to (a^*, c, b, d, f, e) is invertible, one also has that if (a, b, c, d, e, f) is not tetrahedral, then neither is (a^*, c, b, d, f, e) . An analogous argument holds for the operation mapping (a, b, c, d, e, f) to (c, e^*, d^*, f, b, a^*) with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

Remark 11. The group G of order 48 appearing in Proposition 10 corresponds to the subgroup of all possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple (A, B, C, D, E, F) which preserves the set of linear constraints in Eq. (14). The group G is isomorphic to both the direct product $S_2 \times S_4$ and to the signed permutation group of three elements, i.e., $(S_2 \times S_2 \times S_2) \rtimes S_3$.

The first isomorphism, $G \simeq S_2 \times S_4$, can be understood as S_2 acting by the inversion of all eigenvalues,

$$(a, b, c, d, e, f) \leftrightarrow (a^*, b^*, c^*, d^*, e^*, f^*), \tag{28}$$

while S_4 acts by permutation of the *vertices* of the tetrahedron in Fig. 1: each vertex permutation induces a permutation of the six directed edges with the possibility of inverting the orientation of a subset of the edges (such edge orientation reversals can be accounted for a corresponding inversion of eigenvalues).

The second isomorphism, $G \simeq (S_2 \times S_2 \times S_2) \rtimes S_3$, can be understood by considering the triplet $(A, D), (B, E), (C, F)$ of non-adjacent pairs of edges in Fig. 1. Accordingly, it is helpful to arrange the sextuple (a, b, c, d, e, f) into a 2×3 grid using the following identification:

$$(a \ b \ c \ d \ e \ f) \cong \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}. \tag{29}$$

Then one can see that each S_2 factor in the direct product $S_2 \times S_2 \times S_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} a & e^* & f^* \\ d & b^* & c^* \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} a & b & f \\ d & e & c \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} d^* & e & c \\ a^* & b & f \end{pmatrix}. \tag{30}$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{pmatrix} b & c^* & a^* \\ e & f & d \end{pmatrix} \quad \begin{pmatrix} c^* & a & b^* \\ f & d & e \end{pmatrix} \\ \begin{pmatrix} a^* & c & b \\ d & f & e \end{pmatrix} \quad \begin{pmatrix} b^* & a^* & c^* \\ e & d & f \end{pmatrix} \quad \begin{pmatrix} c & b^* & a \\ f & e & d \end{pmatrix} \tag{31}$$

We further remark that these symmetries also corresponding to symmetries of the Wigner $6j$ operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ in Definition 39 with the involution $\lambda \mapsto \lambda^*$ corresponding taking the dual representation $V_\lambda^n \mapsto (V_\lambda^n)^*$. In the case where $n = 2$, the involution map becomes the identity map (as every irreducible representation of $SU(2)$ is self-dual, or the reversal of a pair $(x, -x)$ of traceless eigenvalues is trivial), and thus the symmetry group collapses to a group of order $|S_4| = 4! = 24$. prop:shifting

Proposition 12 (Shifting symmetry). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Define shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ by*

$$\begin{aligned} \tilde{a}_i &:= a_i + x, & \tilde{b}_i &:= b_i + y, & \tilde{c}_i &:= c_i + x + y, \\ \tilde{d}_i &:= d_i + z, & \tilde{e}_i &:= e_i + x + y + z, & \tilde{f}_i &:= f_i + y + z. \end{aligned} \tag{32}$$

Then shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ are tetrahedral if and only if the unshifted eigenvalues (a, b, c, d, e, f) are tetrahedral.

Proof. Given a Hermitian matrix A with eigenvalues $\text{eig}(A) = a$ and a constant $w \in \mathbb{R}$, the eigenvalues of $A + wI$ (where I is the identity matrix) are

$$\text{eig}(A + wI) = (a_1 + w \geq \dots \geq a_n + w). \quad (33)$$

The statement then follows because the sextuple of matrices $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$, where

$$\begin{aligned} \tilde{A} &:= A + xI, & \tilde{B} &:= B + yI, & \tilde{C} &:= C + (x + y)I, \\ \tilde{D} &:= D + zI, & \tilde{E} &:= E + (x + y + z)I, & \tilde{F} &:= F + (y + z)I, \end{aligned} \quad (34)$$

have the shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ and satisfy the tetrahedral constraints in Eq. (14) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (14). $\square$

As a corollary of the shifting symmetries (Proposition 12) one concludes that to solve the tetrahedral Horn problem, in a certain sense, it suffices to solve the tetrahedral Horn problem merely for traceless matrices (Corollary 13) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 14). cor:traceless

Corollary 13 (Traceless eigenvalues). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and for any list of n eigenvalues $x = (x_1 \geq \dots \geq x_n)$, let $x^0 := x - \text{Tr}[x]/n$ denote the traceless list of eigenvalues obtained by shifting x by $-\text{Tr}[x]/n$. Then (a, b, c, d, e, f) are tetrahedral if and only if (i) their traces $(\text{Tr}[a], \text{Tr}[b], \text{Tr}[c], \text{Tr}[d], \text{Tr}[e], \text{Tr}[f])$ are tetrahedral (they satisfy Eq. (19)) and (ii) $(a^0, b^0, c^0, d^0, e^0, f^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (19)) and applications of the shifting symmetry Proposition 12 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (35)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn problem it suffices to solve the tetrahedral Horn problem for non-negative (or positive) eigenvalues only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 12 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure non-negative (or positive) eigenvalues for all six Hermitian matrices. cor:psd

Corollary 14 (Non-negativity). *If (a, b, c, d, e, f) are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$c_n \geq a_n + b_n, \quad f_n \geq b_n + d_n, \quad e_n \geq d_n + c_n, \quad e_n \geq a_n + f_n. \quad (36) \quad \text{\textcolor{blue}{eq:min_eig_ineqs}}$$

Moreover, if (a, b, c, d, e, f) are a sextuple of candidate eigenvalues satisfying Eq. (36), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying eq:positive_shift

$$x \geq -a_n, \quad y \geq -b_n, \quad z \geq -d_n, \quad (37)$$

the resulting shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ (given by Eq. (32)) are all non-negative.

Proof. That tetrahedral eigenvalues satisfy Eq. (36) follows from four applications of the well-known inequality $c_n \geq a_n + b_n$ for the original Horn problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue x_n of a matrix X is characterized by a minimization: $x_n = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$c_n = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = a_n + b_n. \quad (38)$$

Four applications of this inequality produces Eq. (36). Now if (x, y, z) satisfy Eq. (37), we have for all non-negativity of the shifted eigenvalues, e.g.,

$$\forall j : \quad \tilde{a}_j = a_j + x \geq a_n + x \geq 0, \quad (39)$$

Similar statements hold for b, c, d, e and f . $\square$

As a consequence of Corollary 14, we will, without loss of generality, henceforth only consider candidate eigenvalues (a, b, c, d, e, f) which are non-negative in the sense that they are associated to positive semidefinite Hermitian matrices.

Remark 15. Although this section listed a handful of symmetries of the tetrahedral Horn problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case of 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, d, e, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (40)$$

Combining this symmetry with the permutational symmetries given by Proposition 10 (noting eigenvalues of traceless 2×2 Hermitian matrices are invariant under negation) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (41)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (42)$$

Remark 16. The Regge symmetry of a tetrahedron can be understood as a consequence of an earlier discovery of a surprising symmetry of Wigner $6j$ symbol for $SU(2)$ originally due to Regge [Reg59]. The paper due to Boalch provides a comprehensive and modern treatment of the Regge symmetry for the $6j$ symbol for $SU(2)$ by viewing it as a consequence of a symmetry of multiplicity-free $6j$ -symbols for $SU(3)$ [Boa07]. That the Regge symmetry of a tetrahedron emerges from the Regge symmetry of the associated $6j$ symbols can be viewed as a consequence of the semiclassical limit of the latter [PR68; Rob99]. It has also been shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dhen invariants [AI19]. At the moment, it remains unclear if analogous symmetries for the generalized Wigner $6j$ symbol exist for $n \geq 3$ (although there are conjectured symmetries for the $6j$ symbols for $U_q(\mathfrak{sl}_n)$ which would follow from the eigenvalue hypothesis [MS18; LS23]). It could also be possible for additional symmetries to be present in the tetrahedral Horn problem for $n \geq 3$ which are *not* present in the $6j$ symbol for $U(n)$.

3 Background material

sec:background

In this section, we review material on the representation theory of compact connected Lie groups, specifically $U(n)$, some probability theory and also the polynomial functions $\varphi_\lambda(X)$, $\varphi_\gamma^{\alpha\beta}(X, Y)$ and $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ of one, two and three $n \times n$ matrices respectively.

3.1 Representation theory

Even though we will primarily deal with $G = U(n)$, we start with a more general story.

3.1.1 G compact

Let G be a compact connected Lie group, $T \subset G$ a maximal torus, $\mathfrak{t} = \text{Lie}(T)$ its Lie algebra, $\Lambda = \ker(\exp) \subset \mathfrak{t}$ the integral lattice, $\Lambda^* \subset \mathfrak{t}^*$ the weight lattice, and $\Lambda_+^* \subset \Lambda^*$ the set of dominant weights of G . For $\lambda \in \Lambda^*$, we denote by $\pi_\lambda : G \rightarrow \text{End}(V_\lambda)$ the corresponding irreducible representation of G . We will always assume that the space V_λ is equipped with an invariant Hermitian structure such that π_λ becomes a unitary representation. All representations π_λ extend to the corresponding complex Lie group $G^\mathbb{C}$.

The tensor product of two irreducible representations admits a unique decomposition of the form

$$V_\alpha \otimes V_\beta = \bigoplus_{\gamma} C_{\alpha\beta}^\gamma \otimes V_\gamma, \quad \text{\texttt{\{eq:tensor product\}}} \quad (43)$$

where $C_{\alpha\beta}^\gamma$ are multiplicity spaces with dimensions $c_{\alpha\beta}^\gamma = \dim C_{\alpha\beta}^\gamma$. Hermitian structures on V_λ 's induces Hermitian structures on the multiplicity spaces $C_{\alpha\beta}^\gamma$. In particular, the decomposition (43) is an orthogonal decomposition. We will denote the corresponding orthogonal projectors by $P_{\alpha\beta}^\gamma$. Note that

$$\pi_\alpha(g) \otimes \pi_\beta(g) = \sum_{\gamma} (\text{id} \otimes \pi_\gamma(g)) P_{\alpha\beta}^\gamma, \quad \text{\texttt{\{eq:operator product\}}} \quad (44)$$

where on the left hand side we use the tensor product $V_\alpha \otimes V_\beta$, and on the right hand side $C_{\alpha\beta}^\gamma \otimes V_\gamma$.

Characters of irreducible representations are denoted by

$$\chi_\lambda(g) = \text{Tr}_{V_\lambda}[\pi_\lambda(g)]. \quad \text{\texttt{\{eq:character\}}} \quad (45)$$

Taking a trace of equation (44) we obtain the standard product rule for characters:

$$\chi_\alpha(g) \chi_\beta(g) = \sum_{\gamma} c_{\alpha\beta}^\gamma \chi_\gamma(g). \quad \text{\texttt{\{eq:product rule\}}} \quad (46)$$

Triple tensor products can be decomposed in two alternative ways:

$$\begin{aligned} (V_\alpha \otimes V_\beta) \otimes V_\delta &= \bigoplus_{\gamma, \epsilon} (C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon) \otimes V_\epsilon, \\ V_\alpha \otimes (V_\beta \otimes V_\delta) &= \bigoplus_{\phi, \epsilon} (C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi) \otimes V_\epsilon. \end{aligned} \quad (47)$$

We will denote the corresponding orthogonal projectors by $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ and $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$, respectively (the letters γ and ϕ will always be associated to these decompositions). For fixed α, β, δ and γ, ϕ, ϵ , consider the composition of these two projectors $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ and $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. It induces a G -equivariant linear map:

$$(C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon) \otimes V_\epsilon \rightarrow (C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi) \otimes V_\epsilon. \quad (48)$$

Since π_ϵ is an irreducible representation, by Schur's Lemma this map is a tensor product of the identity on V_ϵ , and a map

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon \rightarrow C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi \quad (49)$$

called Wigner's $6j$ -symbol.

Recall that for Hermitian vector spaces V_1, V_2 and an operator $O : V_1 \rightarrow V_2$ one can introduce the norms

$$\begin{aligned} \|O\|_\infty &= \max_{v \in V_1, \|v\|_{V_1}=1} \|O(v)\|_{V_2}, \\ \|O\|_2 &= \text{Tr}_{V_1}[O^\dagger O]^{\frac{1}{2}} = \text{Tr}_{V_2}[OO^\dagger]^{\frac{1}{2}}, \end{aligned} \quad (50)$$

which satisfy the following estimates:

$$\|O\|_\infty^2 \leq \|O\|_2^2 \leq \dim(V_1) \|O\|_\infty^2. \quad \text{\texttt{\{eq:norm estimates\}}} \quad (51)$$

In particular, for $6j$ -symbols, we have

$$\left\| \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \right\|_\infty^2 = \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} \right\|_\infty^2 \leq \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} \right\|_2^2 = \frac{1}{\dim(V_\epsilon)} \text{Tr}(\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma}). \quad (52)$$

In the above formula, we have used the fact that for two orthogonal Hermitian projectors Π_1 and Π_2 one has

$$\|\Pi_1 \Pi_2\|_2^2 = \text{Tr}((\Pi_1 \Pi_2)(\Pi_1 \Pi_2)^\dagger) = \text{Tr}(\Pi_1 \Pi_2^2 \Pi_1) = \text{Tr}(\Pi_1 \Pi_2). \quad (53)$$

For future use, we will need the following fact:

prop: Pi Omega Pi

Proposition 17. *Let Π_1, Π_2 be two orthogonal projections, and Ω be a non-negative Hermitian operator. Then,*

$$|\text{Tr}(\Pi_1 \Omega \Pi_2)| \leq \|\Pi_1 \Pi_2\|_\infty (\text{Tr } \Pi_1 \Omega)^{\frac{1}{2}} (\text{Tr } \Omega \Pi_2)^{\frac{1}{2}}. \quad (\text{eq: Pi Omega Pi}) \quad (54)$$

Proof. We compute,

$$\begin{aligned} |\text{Tr}(\Pi_1 \Omega \Pi_2)| &= |\text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_1)| \\ &= |\text{Tr}(\Pi_1 \Pi_2 \Omega^{\frac{1}{2}})(\Omega^{\frac{1}{2}} \Pi_1)| \\ &\leq (\text{Tr } \Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1)^{\frac{1}{2}} (\text{Tr } \Pi_1 \Omega \Pi_1)^{\frac{1}{2}} \\ &= (\text{Tr } \Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1)^{\frac{1}{2}} (\text{Tr } \Pi_1 \Omega)^{\frac{1}{2}}, \end{aligned} \quad (55)$$

where in the third line we have used the Cauchy inequality for the Hilbert-Schmidt norm. Let $U = \text{im}(\Pi_2)$ and choose a basis of eigenvectors $\{e_j\}$ of the operator $\Pi_2 \Omega \Pi_2$ with eigenvalues $\omega_j \geq 0$. Then,

$$\begin{aligned} \text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1) &= \text{Tr}(\Pi_2 \Pi_1^2 \Pi_2)(\Pi_2 \Omega \Pi_2) \\ &= \sum_j \omega_j \|\Pi_1 \Pi_2 e_j\|^2 \\ &\leq \|\Pi_1 \Pi_2\|_\infty^2 \text{Tr}(\Pi_2 \Omega \Pi_2) \\ &= \|\Pi_1 \Pi_2\|_\infty^2 \text{Tr}(\Omega \Pi_2). \end{aligned} \quad (56)$$

This completes the proof. $\square$

3.1.2 $G = U(n)$ and the function φ_λ

For $G = U(n)$, dominant weights are parametrized by non-increasing arrays of integers

$$\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n). \quad (57)$$

In particular, the first fundamental weight $\varpi_1 = (1, 0, \dots, 0)$ corresponds to the defining representation of $U(n)$ on $V_{\varpi_1} = \mathbb{C}^n$.

We say that a dominant weight λ is positive if $\lambda_n \geq 0$, and we denote the set of positive weights by Λ_{++}^* . We will use the notation

$$|\lambda| = \sum_{i=1}^n \lambda_i, \quad l(\lambda) = |\{i; \lambda_i \neq 0\}|, \quad (58)$$

where $l(\lambda)$ is called the depth of λ . Positive weights λ can be viewed as partitions of $|\lambda|$ into $l \leq n$ parts, or as Young diagrams with $|\lambda|$ boxes and at most n rows. Representations $\pi_\lambda(g)$ corresponding to positive weights are polynomial in matrix elements of g , and they extend to matrix algebras $\text{Mat}_n(\mathbb{C})$. That is,

$$\pi_\lambda(X) \pi_\lambda(Y) = \pi_\lambda(XY) \quad (59)$$

for all $X, Y \in \text{Mat}_n(\mathbb{C})$. Tensor products of representations with positive weights decompose into direct sums of representations of the same type. Note that in the decomposition (43) one has only terms with

$$|\gamma| = |\alpha| + |\beta|. \quad (\text{eq:trace representations}) \quad (60)$$

We will sometimes refer to equation (60) as to a trace equality (in fact, one of many trace equalities in this paper).

Recall that for Young diagrams (or partitions) λ with fixed size $|\alpha| = k$ label isomorphism classes of irreducible representations W_λ of the symmetric group S_k . The dimensions of these representations are given by formula

$$\dim(W_\lambda) = \frac{|\lambda|!}{H_\lambda}, \quad H_\lambda = \prod_{b \in \lambda} h_\lambda(b), \quad (\text{eq:dim_W_hook_product}) \quad (61)$$

where the product is over the boxes in the Young diagram associated to λ , and $h_\lambda(b)$ is hook length of b which is equal to the number of boxes in the hook with the corner at b , and with the arms extending to the right of b and downwards from b . For the future use, we introduce rescaled operators

$$\Phi_\lambda(X) = \frac{1}{H_\lambda} \pi_\lambda(X), \quad (62)$$

and rescaled characters

$$\varphi_\lambda(X) = \text{Tr}_{V_\lambda} \Phi_\lambda(X) = \frac{1}{H_\lambda} \chi_\lambda(X). \quad (63)$$

Note that

$$\sum_{\lambda: |\lambda|=k} \varphi_\lambda(X) = \frac{1}{k!} [\text{Tr}(X)]^k, \quad \sum_{\lambda} \varphi_\lambda(X) = \exp(\text{Tr}(X)). \quad (64) \quad \text{\textcolor{blue}{\code{eq:sum lambda}}}$$

Indeed, by the Schur-Weyl duality, there is an isomorphism of $(S_k \otimes U(n))$ -modules

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{|\lambda|=k} W_\lambda \otimes V_\lambda, \quad (65) \quad \text{\textcolor{blue}{\code{eq:schur-weyl}}}$$

where the symmetric group S_k acts by permutations of tensor factors in $(\mathbb{C}^n)^{\otimes k}$. Since all $U(n)$ representations entering the decomposition (65) are positive, the Schur-Weyl duality extends to an isomorphism of modules for monoids $(S_k \otimes \text{Mat}_n(\mathbb{C}))$. Applying it to an element $X \in \text{Mat}_n(\mathbb{C})$, we obtain

$$X^{\otimes k} = \bigoplus_{|\lambda|=k} \text{Id}_{W_\lambda} \otimes \pi_\lambda(X), \quad (66) \quad \text{\textcolor{blue}{\code{eq:X^k}}}$$

where on the right hand side we have a direct sum of operators acting on the subspaces of the Schur-Weyl decomposition. Taking a trace of this equation over $(\mathbb{C}^n)^{\otimes k}$ yields

$$[\text{Tr}(X)]^k = \text{Tr}(X^{\otimes k}) = \sum_{\lambda: |\lambda|=k} \dim(W_\lambda) \chi_\lambda(X) = k! \sum_{\lambda: |\lambda|=k} H_\lambda^{-1} \chi_\lambda(X) = k! \sum_{\lambda: |\lambda|=k} \varphi_\lambda(X), \quad (67)$$

as required. The second identity in (64) follows by taking a sum over k in the first one.

Proposition 18. *For $X \in \mathcal{H}_n^+$, $\lambda \in \Lambda_{++}^*$, operators $\pi_\lambda(X)$ and $\Phi_\lambda(X)$ are non-negative Hermitian operators, and functions $\varphi_\lambda(X)$ take non-negative real values on $\mathcal{H}_n^+$.*

Proof. By (66), operators $\text{id}_{W_\lambda} \otimes \pi_\lambda(X)$ are orthogonal restrictions of the non-negative Hermitian operator $X^{\otimes k}$. Hence, they are also non-negative Hermitian operators, and so are $\pi_\lambda(X)$ and $\Phi_\lambda(X)$. Since $\varphi_\lambda(X)$ is the trace of $\Phi_\lambda(X)$, it takes values in $\mathbb{R}_{\geq 0}$. $\square$

3.1.3 Functions $\varphi_\gamma^{\alpha\beta}$ and $\varphi_{\gamma \in \phi}^{\alpha\beta\delta}$

It is convenient to introduce the following functions of two matrices,

$$\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}_{V_\alpha \otimes V_\beta} (\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma. \quad (68)$$

Proposition 19. *For $\alpha, \beta \in \Lambda_{++}^*$ and $X, Y \in \text{Mat}_n(\mathbb{C})$, we have*

$$\varphi_\alpha(X) \varphi_\beta(Y) = \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y). \quad (69) \quad \text{\textcolor{blue}{\code{eq:product varphi}}}$$

For $X, Y \in \mathcal{H}_n^+$, the function $\varphi_\gamma^{\alpha\beta}(X, Y)$ takes non-negative real values.

Proof. For the first statement, we compute

$$\sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y) = \sum_{\gamma} \text{Tr}_{V_\alpha \otimes V_\beta} (\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma = \text{Tr}_{V_\alpha \otimes V_\beta} (\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) = \varphi_\alpha(X) \varphi_\beta(Y). \quad (70)$$

Here we have used the fact that $\sum_{\gamma} P_{\alpha\beta}^\gamma = \text{id}_{V_\alpha \otimes V_\beta}$. For the second statement, we have

$$\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}_{V_\alpha \otimes V_\beta} (\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma = \text{Tr}_{V_\alpha \otimes V_\beta} P_{\alpha\beta}^\gamma (\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma. \quad (71)$$

Hence, $\varphi_\gamma^{\alpha\beta}(X, Y)$ is the trace of the orthogonal restriction of $(\Phi_\alpha(X) \otimes \Phi_\alpha(Y))$ to $(C_{\alpha\beta} \otimes V_\gamma) \subset V_\alpha \otimes V_\beta$. Since $(\Phi_\alpha(X) \otimes \Phi_\alpha(Y))$ is a non-negative Hermitian operator, so is its restriction. It has non-negative real eigenvalues, and a non-negative real trace, as required. $\square$

Theorem 20. For $X, Y \in \text{Mat}_n(\mathbb{C})$ and $\gamma \in \Lambda_{++}^*$, we have

$$\varphi_\gamma(X + Y) = \sum_{\alpha, \beta} \varphi_\gamma^{\alpha\beta}(X, Y). \quad (72)$$

Proof. Let $k = |\gamma|$, ρ_k the natural representation of the monoid $(S_k \times \text{Mat}_n(\mathbb{C}))$ on the space $C_k = (\mathbb{C}^n)^{\otimes k}$, and P^γ the orthogonal projector to the summand $W_\gamma \otimes V_\gamma$ in the Schur-Weyl decomposition (65). We compute,

$$\begin{aligned} \varphi_\gamma(X + Y) &= h_\gamma^{-1} \text{Tr}_{V_\lambda} \pi_\gamma(X + Y) \\ &= (h_\gamma d_\gamma)^{-1} \text{Tr}_{C_k} P^\gamma \rho_k(X + Y) \\ &= k!^{-1} \text{Tr}_{C_k} P^\gamma (X + Y)^{\otimes k} \\ &= k!^{-1} \text{Tr}_{C_k} P^\gamma \sum_{l=0}^k \sum_{\sigma \in S_k} \frac{k!}{l!(k-l)!} \rho_k(\sigma)(X^{\otimes l} \otimes Y^{\otimes(k-l)}) \rho_k(\sigma^{-1}) \\ &= \sum_{\alpha, \beta} (|\alpha|! |\beta|!)^{-1} \text{Tr}_{(W_\alpha \otimes V_\alpha) \otimes (W_\beta \otimes V_\beta)} P_{\alpha\beta}^\gamma \pi_\alpha(X) \otimes \pi_\beta(Y) \\ &= \sum_{\alpha, \beta} (h_\alpha h_\beta)^{-1} \text{Tr}_{V_\alpha \otimes V_\beta} P_{\alpha\beta}^\gamma (\pi_\alpha(X) \otimes \pi_\beta(Y)) \\ &= \sum_{\alpha, \beta} \text{Tr}_{V_\alpha \otimes V_\beta} P_{\alpha\beta}^\gamma (\varphi_\alpha(X) \otimes \varphi_\beta(Y)) \\ &= \sum_{\alpha, \beta} \phi_{\alpha\beta}^\gamma(X, Y). \end{aligned} \quad (73)$$

Here in the fourth line we have used the fact that

$$(X + Y)^{\otimes k} = \sum_{l=0}^k \frac{k!}{l!(k-l)!} \sum_{\sigma \in S_k} \rho_k(\sigma)(X^{\otimes l} \otimes Y^{\otimes(k-l)}) \rho_k(\sigma^{-1}) \quad (74)$$

which is a non-commutative counterpart of the binomial decomposition. In the fifth line, we have used the Schur-Weyl duality for representations $_{k\rho_l}$ and $_{\rho_{k-l}}$, and the fact that $P^\gamma = \sum_{\alpha, \beta} P_{\alpha\beta}^\gamma$. $\square$

Remark 21. While equation (69) is a generalization of the product rule for characters (46), equation (72) uses the Schur-Weyl duality, and it is a specific feature of $G = U(n)$, and of the corresponding matrix algebra $\text{Mat}_n(\mathbb{C})$.

For a 6-tuple of positive weights, we introduce a function of three matrices,

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}_{V_\alpha \otimes V_\beta \otimes V_\delta} \Pi_{\alpha\beta\delta}^{\epsilon; \phi}(\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)) \Pi_{\alpha\beta\delta}^{\epsilon; \gamma}. \quad (75)$$

Note that the following trace equalities are necessary conditions for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}$ being non-zero:

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\phi| = |\epsilon|, \quad |\gamma| + |\delta| = |\epsilon|. \quad (76)$$

Even when restricted to $(\mathcal{H}_n^+)^{\times 3}$, functions $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}$ take complex values, in general. However, the following is true:

Proposition 22. For $X, Y, Z \in \mathcal{H}_n^+$, the functions

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \sum_{\gamma} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \quad \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) = \sum_{\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \quad (77)$$

take non-negative real values.

Proof. We compute,

$$\begin{aligned} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\gamma} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \\ &= \sum_{\gamma} \text{Tr}_{V_\alpha \otimes V_\beta \otimes V_\delta} \Pi_{\alpha\beta\delta}^{\epsilon; \phi}(\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)) \Pi_{\alpha\beta\delta}^{\epsilon; \gamma} \\ &= \text{Tr}_{V_\alpha \otimes V_\beta \otimes V_\delta} \Pi_{\alpha\beta\delta}^{\epsilon; \phi}(\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)). \end{aligned} \quad (78)$$

Here we have used the fact that $\sum_{\gamma} \Pi_{\alpha\beta\delta}^{\epsilon; \gamma} = P_{\alpha\beta\delta}^\epsilon$, where $P_{\alpha\beta\delta}^\epsilon$ is the orthogonal projector to the isotypical component of ϵ in the triple tensor product. We have also used the fact that $P_{\alpha\beta\delta}^\epsilon \Pi_{\alpha\beta\delta}^{\epsilon; \phi} = \Pi_{\alpha\beta\delta}^{\epsilon; \phi}$. The right hand side in the equation above is the trace of the orthogonal restriction of the non-negative Hermitian operator $(\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z))$ to the image of the projector $\Pi_{\alpha\beta\delta}^{\epsilon; \phi}$. Hence, it is a non-negative real number. The proof for functions $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ is similar. $\square$

3.2 Probability measures

In this Section, we collect information about probability measures on finite sets and on conjugacy classes of Hermitian matrices needed later in the paper.

3.2.1 Measures on finite sets

Recall that for a finite set $[r] = \{1, \dots, r\}$ a probability measure is defined by an array of non-negative real numbers $(p_1, \dots, p_r)$ such that $\sum_{i=1}^r p_i = 1$. For two measures p and q on $[r]$, there are several ways estimate how close they are. First, there are standard distances in $\mathbb{R}^n$:

$$K(p, q) := \frac{1}{2} \|p - q\|_1 = \frac{1}{2} \sum_{i=1}^r |p_i - q_i|, \quad \|p - q\|_2^2 = \sum_{i=1}^r (p_i - q_i)^2, \quad (79)$$

where $K(p, q)$ is the Kolmogorov distance. As usual, each of these norms vanishes if and only if $p = q$. Another measure of similarity between distributions p and q is the Bhattacharyya coefficient [Bha46; FV02]:

$$\text{BC}(p, q) = \sum_{i=1}^r \sqrt{p_i q_i}. \quad \text{\texttt{\textcolor{blue}{eq:Bhattacharyya_coefficient}}} \quad (80)$$

For all distributions p, q , the Bhattacharyya coefficient $\text{BC}(p, q)$ lies in the interval $[0, 1]$, and $\text{BC}(p, q) = 1$ if and only if $p = q$. Recall the following relation between the Kolmogorov distance and the Bhattacharyya coefficient (see *e.g.* [FV02, Proposition 5, Equation (41)]):

$$1 - \text{BC}(p, q) \leq K(p, q) \leq \sqrt{1 - \text{BC}(p, q)^2}. \quad (81)$$

This implies⁴

$$\text{BC}(p, q) \leq \sqrt{1 - K(p, q)^2} \leq \exp\left(-\frac{1}{2} K(p, q)^2\right) = \exp\left(-\frac{1}{8} \|p - q\|_1^2\right). \quad \text{\texttt{\textcolor{blue}{eq:BC norm estimate}}} \quad (82)$$

We extend the Bhattacharyya coefficient to non-normalized non-negative tuples with $\sum_i p_i = \sum_i q_i = \tau > 0$. In that case, the *unnormalized* Bhattacharyya coefficient is also denoted by

$$\text{BC}(p, q) = \tau B\left(\frac{p}{\tau}, \frac{q}{\tau}\right), \quad (83)$$

and the estimate (82) then reads

$$\frac{\text{BC}(p, q)}{\tau} \leq \exp\left(-\frac{1}{8\tau^2} \|p - q\|_1^2\right). \quad \text{\texttt{\textcolor{blue}{eq:bhattacharyya_vs_distance}}} \quad (84)$$

Finally, the Kullback-Leibler (KL) divergence is defined as

$$D(p||q) = \sum_{i=1}^n p_i \log \frac{p_i}{q_i}. \quad \text{\texttt{\textcolor{blue}{eq:KL divergence}}} \quad (85)$$

It takes values between 0 and $+\infty$, and it is equal to zero if and only if $p = q$.

3.2.2 Measures on partitions

We define the set

$$\mathbb{R}_n^+ = \{x = (x_1 \geq x_2 \geq \dots \geq x_n) \in \mathbb{R}^n; x \neq 0, x_n \geq 0\}. \quad (86)$$

One can associate to $x \in \mathbb{R}_n^+$ a probability measure p^x on the set $[n]$ by assigning

$$p_i^x = \frac{x_i}{|x|}, \quad \text{where} \quad |x| = \sum_{i=1}^n x_i. \quad (87)$$

⁴Here we used the inequality $\sqrt{1 - x^2} \leq \exp(-x^2/2)$ which holds for all $x \in [0, 1]$.

Let $X = \text{diag}(x_1, \dots, x_n)$ be a diagonal non-negative Hermitian matrix with x_i 's on the diagonal. By abuse of notation, we write $\varphi_\lambda(x) = \varphi_\lambda(X)$. For each x and each $k \geq 1$, one can define a probability measure on the set of positive weights of $U(n)$ with $|\lambda| = k$:

$$P_x(\lambda) = \frac{k!}{|x|^k} \varphi_\lambda(x). \quad \text{\textcolor{blue}{\{eq:P_x(lambda)\}}} \quad (88)$$

The following result is known as eigenvalue or spectrum estimation [ARS88; KW01; HM02; CM06; OW16]: \textcolor{blue}{prop:eig_est}

Proposition 23 (Eigenvalue estimation). *For, $x \in \mathbb{R}_+^n, \lambda \in \Lambda_{+++}^*$, we have*

$$P_x(\lambda) \leq (k+1)^{\binom{n}{2}} \exp(-kD(p^\lambda \| p^x)). \quad (89)$$

Proposition 53 shows that for k large the measure P_x is concentrated on positive weights λ which have a shape close to the one of $x \in \mathbb{R}_+^n$. The following proposition compares measures P_x and P_y for different $x, y \in \mathbb{R}_+^n$.

Proposition 24. *For $k \geq 1$ and $x, y \in \mathbb{R}_+^n$ with $|x| = |y| = \tau$, one has*

$$\text{BC}(P_x, P_y) \leq (k+1)^{\binom{n}{2}} \text{BC}(p^x, p^y)^k \quad (90)$$

Proof. First, we estimate

$$P_x(\lambda) = \frac{k!}{|x|^k H_\lambda} \chi_\lambda(x) \leq \frac{k!}{|x|^k} \dim(V_\lambda) \prod_{i=1}^n \frac{x_i^{\lambda_i}}{\lambda_i!} \leq \frac{k!}{|x|^k} (k+1)^{\binom{n}{2}} \prod_{i=1}^n \frac{x_i^{\lambda_i}}{\lambda_i!}. \quad (91)$$

Here we have used that $H_\lambda \geq \prod_{i=1}^n \lambda_i!$, that $\prod_{i=1}^n x_i^{\lambda_i}$ is the maximal monomial in the Schur-polynomial $\chi_\lambda(x)$, and the estimate \textcolor{blue}{\{eq:unitary_irrep_dim_bound\}}

$$\dim V_\lambda^n \leq (k+1)^{\binom{n}{2}}. \quad (92)$$

Next, we compute

$$\text{BC}(P_x, P_y) = \sum_{\lambda: |\lambda|=k} (P_x(\lambda) P_y(\lambda))^{\frac{1}{2}} \leq \frac{k!}{|x|^k} (k+1)^{\frac{n(n-1)}{2}} \sum_{\lambda: |\lambda|=k} \prod_{i=1}^n \frac{(x_i y_i)^{\frac{\lambda_i}{2}}}{\lambda_i!} \leq (k+1)^{\binom{n}{2}} \text{BC}(p^x, p^y)^k. \quad (93)$$

Here in the last inequality we have replaced the sum over positive weight with $\lambda: |\lambda| = k$ by a sum over all arrays $(\lambda_1, \dots, \lambda_n)$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = k$ which allows to use the binomial formula. $\square$

3.2.3 Measures on conjugacy classes

For $x \in \mathbb{R}_+^n$, we define the conjugacy class $\mathcal{O}_x$ of the diagonal matrix $[x] = \text{diag}(x_1, \dots, x_n)$ under the action of the unitary group $U(n)$:

$$\mathcal{O}_x = \{X = u[x]u^{-1}; u \in U(n)\} \subset \mathcal{H}_n^+. \quad (94)$$

Each conjugacy class carries a unique $U(n)$ -invariant probability measure μ_x . This measure can be constructed as the pushforward of the probability Haar measure on $U(n)$. Also, conjugacy classes are isomorphic to coadjoint orbits of $U(n)$ and they carry canonical Duistermaat-Heckman measure proportional to μ_x . In what follows, we will need expectation values $\mathbb{E}_x$ on conjugacy classes $\mathcal{O}_x$ with respect to the measure μ_x , $\mathbb{E}_{x,y}$ on $\mathcal{O}_x \times \mathcal{O}_y$ with the measure $\mu_x \times \mu_y$, and $\mathbb{E}_{x,y,z}$ on $\mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z$ with the measure $\mu_x \times \mu_y \times \mu_z$. \textcolor{blue}{prop: estimate E_{x,y,z}}

Proposition 25. *We have,*

$$\begin{aligned} \mathbb{E}_x \Phi_\lambda(X) &= \frac{\varphi_\lambda(x)}{\dim V_\lambda} \text{id}_{V_\lambda}, \\ \mathbb{E}_{x,y} \frac{\varphi_\gamma^{\alpha\beta}(X,Y)}{\dim V_\gamma} &= c_{\alpha\beta}^\gamma \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta}, \\ \mathbb{E}_{x,y,z} \frac{\varphi_{\gamma\epsilon\delta}^{\alpha\beta\delta}(X,Y,Z)}{\dim V_\epsilon} &= \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{\varphi_\delta(y)}{\dim V_\delta}. \end{aligned} \quad (95)$$

Proof. For the first equality, by Schur's Lemma we have

$$\mathbb{E}_x \Phi_\lambda(X) = \int_{O_x} \Phi_\lambda(X) d\mu_x = c(x) \text{Id}_{V_\lambda} \quad (96)$$

for some $c(x) \in \mathbb{C}$. By taking a trace of the equality above, we obtain

$$\varphi_\lambda(x) = c(x) \dim V_\lambda \Rightarrow c(x) = \frac{\varphi_\lambda(x)}{\dim V_\lambda}, \quad (97)$$

as required. For the second equality, we compute

$$\begin{aligned} \mathbb{E}_{x,y} \frac{\varphi_\gamma^{\alpha\beta}(X,Y)}{\dim V_\gamma} &= \frac{1}{\dim V_\gamma} \mathbb{E}_{x,y} \text{Tr}(\Phi_\alpha(X) \otimes \Phi_\beta(Y)) P_{\alpha\beta}^\gamma \\ &= \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{1}{\dim V_\gamma} \text{Tr} P_{\alpha\beta}^\gamma \\ &= c_{\alpha\beta}^\gamma \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta}. \end{aligned} \quad (98)$$

Finally, for the third equality we compute

$$\begin{aligned} \mathbb{E}_{x,y,z} \frac{\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X,Y,Z)}{\dim V_\epsilon} &= \frac{1}{\dim V_\epsilon} \mathbb{E}_{x,y,z} \text{Tr} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}(\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)) \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \\ &= \frac{1}{\dim V_\epsilon} \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{\varphi_\delta(y)}{\dim V_\delta} \text{Tr} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \\ &= \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{\varphi_\delta(y)}{\dim V_\delta}. \end{aligned} \quad (99)$$

□

Note that even though functions $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X,Y,Z)$ are complex values, their expectation values are real and positive.

4 Inequalities for tetrahedral eigenvalues

In this Section, we state and prove one of the main results of this paper. To set up notation, we will be considering 6-tuples of non-negative arrays $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ satisfying the trace conditions

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \quad \text{Tr}[b] + \text{Tr}[d] = \text{Tr}[f], \quad \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d] = \text{Tr}[e]. \quad (100)$$

On the space of such 6-tuples, we introduce a function

$$D \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right) := \min_{(A,B,D) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left(\|\text{eig}(A+B) - c\|_1^2 + \|\text{eig}(B+D) - f\|_1^2 + \|\text{eig}(A+B+D) - e\|_1^2 \right). \quad (101)$$

Note that $D \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right) = 0$ if and only if there exists non-negative Hermitian matrices

$$(A, B, C = A + B, D, E = A + B + C, F = B + D) \in (\mathcal{H}_n^+)^6 \quad (102)$$

with ordered eigenvalues given by (a, b, c, d, e, f) . We will also need 6-tuples of positive highest weights

$$(\alpha, \beta, \gamma, \delta, \phi, \epsilon) \in (\Lambda_{++}^*)^6 \quad (103)$$

satisfying the size conditions

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\beta| + |\delta| = |\epsilon|. \quad (104)$$

We denote $|\epsilon| = k$.

thm:iff_tet

Theorem 26. Let $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ be a 6-tuple satisfying the trace conditions. Then, Hermitian matrices (102) with eigenvalues (a, b, c, d, e, f) exist if and only if for all triples of positive weights $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$, the following inequalities are satisfied

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\delta(d)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\epsilon(e))^{\frac{1}{3}}. \quad (105)$$

feq:nec_tet

Furthermore, if inequalities Eq. (105) hold true for all triples $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$ with $|\alpha| + |\beta| + |\delta| = k$, then

$$\frac{1}{\text{Tr}[e]} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq O \left(n \sqrt{\frac{\log k}{k}} \right). \quad (106)$$

feq:tri_abd_cef_dist_bound

We will split the proof of Theorem 26 into several steps.

4.1 Necessary conditions

First, we assume that there are matrices (A, B, C, D, E, F) with eigenvalues (a, b, c, d, e, f) satisfying Eq. (102). We then compute,

$$\begin{aligned} \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &= \varphi_\alpha(A)\varphi_\beta(B)\varphi_\delta(D), \\ &= \sum_{\gamma, \epsilon, \phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D), \\ &\leq \sum_{\gamma, \epsilon, \phi} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \right|, \\ &\leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(A)\varphi_\delta(D)\varphi_\phi(F)\varphi_\gamma(C)\varphi_\epsilon(E))^{\frac{1}{3}}, \\ &\leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\delta(d)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\epsilon(e))^{\frac{1}{3}}. \end{aligned} \quad (107)$$

This calculation follows from the estimate:

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(A)\varphi_\delta(D)\varphi_\phi(F)\varphi_\gamma(C)\varphi_\epsilon(E))^{\frac{1}{3}}. \quad (108)$$

feq:vp_ABD_abs_bound

To prove this inequality, use Proposition 17 to obtain

$$\begin{aligned} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \right| &= \left| \text{Tr } \Pi_{\alpha\beta\delta}^{\epsilon;\phi}(\Phi_\alpha(A) \otimes \Phi_\beta(B) \otimes \Phi_\delta(D)) \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \right| \\ &\leq \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \right\|_\infty (\text{Tr } \Pi_{\alpha\beta\delta}^{\epsilon;\phi}(\Phi_\alpha(A) \otimes \Phi_\beta(B) \otimes \Phi_\delta(D)))^{\frac{1}{2}} \\ &\times (\text{Tr } (\Phi_\alpha(A) \otimes \Phi_\beta(B) \otimes \Phi_\delta(D)) \Pi_{\alpha\beta\delta}^{\epsilon;\gamma})^{\frac{1}{2}} \\ &= \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D)^{\frac{1}{2}} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(A, B, D)^{\frac{1}{2}}. \end{aligned} \quad (109)$$

Then we have the following upper bounds on $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$:

$$\begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) &\leq \sum_{\epsilon\beta\delta} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \\ &= \sum_{\epsilon} \text{Tr } P_{\alpha\beta\delta}^{\epsilon} \left(\Phi_\alpha(A) \otimes \sum_{\beta\delta} \Phi_\phi^{\beta\delta}(B, D) \right) \\ &= \text{Tr } \Phi_\alpha(A) \otimes \Phi_\phi(B + D) \\ &= \varphi_\alpha(A)\varphi_\phi(B + D), \end{aligned} \quad (110)$$

and

$$\begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) &\leq \sum_{\alpha\beta\delta\phi} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \\ &= \sum_{\alpha\phi} \text{Tr } P_{\alpha\beta\delta}^{\epsilon} \left(\Phi_\alpha(A) \otimes \sum_{\beta\delta} \Phi_\phi^{\beta\delta}(B, D) \right) \\ &= \sum_{\alpha\phi} \text{Tr } P_{\alpha\beta\delta}^{\epsilon} (\Phi_\alpha(A) \otimes \Phi_\phi(B + D)) \\ &= \sum_{\alpha\phi} \varphi_{\epsilon}^{\alpha\phi}(A, B + D) \\ &= \varphi_\epsilon(A + B + D). \end{aligned} \quad (111)$$

In both estimates, we are using the fact that $\varphi_{\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \geq 0$. Together, they imply

$$\varphi_{\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \leq (\varphi_\alpha(A)\varphi_\phi(B+D))^{\frac{2}{3}} \varphi_\epsilon(A+B+D)^{\frac{1}{3}} = (\varphi_\alpha(A)\varphi_\phi(F))^{\frac{2}{3}} \varphi_\epsilon(E)^{\frac{1}{3}}. \quad (112)$$

In an analogous fashion, we conclude the following bound on $\varphi_{\gamma\epsilon_-}^{\alpha\beta\delta}(A, B, D)$:

$$\varphi_{\gamma\epsilon_-}^{\alpha\beta\delta}(A, B, D) \leq (\varphi_\gamma(C)\varphi_\delta(D))^{\frac{2}{3}} \varphi_\epsilon(E)^{\frac{1}{3}}. \quad (113)$$

and this completes the proof for Eq. (108).

4.2 Distance bound

Let $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ which satisfy the trace conditions and also

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} \right\|_\infty (\varphi_\alpha(a)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\delta(d)\varphi_\epsilon(e))^{\frac{1}{3}} \quad \{\text{eq:abd_multiplied}\} \quad (114)$$

for all $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$. We will consider triples with $|\alpha| + |\beta| + |\gamma| = k$ for some fixed integer k . Denote by $(\gamma^*, \epsilon^*, \phi^*)$ the triple for which the corresponding term on the right hand side is maximal. We have,

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{3(n-1)} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{matrix} \right\} \right\|_\infty (\varphi_\alpha(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_\delta(d)\varphi_{\epsilon^*}(e))^{\frac{1}{3}}, \quad \{\text{eq:first_abd_bound}\} \quad (115)$$

where $(k+1)^{3(n-1)}$ serves as an upper bound on the number of terms in the sum in (252). Equation (242) then implies

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) > 0 \implies \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{matrix} \right\} \right\|_\infty > 0. \quad (116)$$

Next, consider the bound

$$\begin{aligned} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{matrix} \right\} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &= \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_{\delta^*}^n)}{\dim(V_{\epsilon^*}^n)} \mathbb{E}_{(A, B, D) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \varphi_{\gamma^* \epsilon^* \phi^*}^{\alpha\beta\delta}(A, B, D), \\ &\leq (k+1)^{3\binom{n}{2}} \left| \varphi_{\gamma^* \epsilon^* \phi^*}^{\alpha\beta\delta}(A^*, B^*, D^*) \right| \\ &\leq (k+1)^{3\binom{n}{2}} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{matrix} \right\} \right\|_\infty (\varphi_\alpha(A^*)\varphi_{\phi^*}(F^*)\varphi_{\gamma^*}(C^*)\varphi_\delta(D^*)\varphi_{\epsilon^*}(E^*))^{\frac{1}{3}}. \end{aligned} \quad \{\text{eq:second_abd_bound}\} \quad (117)$$

Here in the first line we have used Proposition 25. In the second line we denote by $(A^*, B^*, D^*) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d$ a triple of matrices which realizes a maximum of the functions $\left| \varphi_{\gamma^* \epsilon^* \phi^*}^{\alpha\beta\delta}(A, B, D) \right|$, and we estimate the dimensions of representations by Eq. (222). In the third line, we denote $C^* = A^* + B^*$, $E^* = A^* + B^* + D^*$, $F^* = B^* + D^*$, and use the estimate in Eq. (108).

By multiplying the estimates (253) and (117), we obtain

$$\begin{aligned} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{matrix} \right\} \right\|_2^2 (\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d))^2 &\leq (k+1)^{3\binom{n}{2}+3n} \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{matrix} \right\} \right\|_\infty^2 \\ &\times (\varphi_\alpha(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_\delta(d)\varphi_{\epsilon^*}(e))^{\frac{1}{3}} \\ &\times (\varphi_\alpha(A^*)\varphi_{\phi^*}(F^*)\varphi_{\gamma^*}(C^*)\varphi_\delta(D^*)\varphi_{\epsilon^*}(E^*))^{\frac{1}{3}}. \end{aligned} \quad (118)$$

This then implies

$$\begin{aligned} \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &\leq (k+1)^{\frac{3}{2}\binom{n}{2}+\frac{3}{2}n} \\ &\times (\varphi_\alpha(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_\delta(d)\varphi_{\epsilon^*}(e))^{\frac{1}{6}} \\ &\times (\varphi_\alpha(A^*)\varphi_{\phi^*}(F^*)\varphi_{\gamma^*}(C^*)\varphi_\delta(D^*)\varphi_{\epsilon^*}(E^*))^{\frac{1}{6}} \end{aligned} \quad (119)$$

Indeed, if $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) = 0$ the inequality is verified since the right hand side is non-negative. Otherwise, if $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) > 0$, then the norms of the $6j$ -symbol are nonvanishing and we can use the standard operator norm estimates from Eq. (51).

TC: COPIED FROM LATER SECTIONS The purpose of this section is to prove our main distance bound, Theorem 2. Note that Theorem 2 then immediately implies the “if” portion of the claim of Theorem 26 by taking $k \rightarrow \infty$. The “only if” portion of Theorem 26 was proven in Corollary 45.

Proof of Theorem 2. The proof of Theorem 2 mimics the structure of the proof of Proposition 59. In the first stage, we fix an arbitrary triple (α, β, δ) with total size $|\alpha| + |\beta| + |\delta| = k$ and derive multiple bounds on the product $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$. Then in the second stage, we sum over all such triples (α, β, δ) and obtain the bound on $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$.

We begin by bounding $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ in two separate ways and then afterward combine them to apply the eigenvalue separation lemma (specifically Eq. (232)). The first bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ is obtained from relaxing the assumption of the claim that (a, b, c, d, e, f) satisfies Eq. (105) for all triples of Young diagrams (α, β, δ) . By multiplying both sides of Eq. (105) by $\varphi_\alpha(a)^{\frac{1}{3}}\varphi_\delta(d)^{\frac{1}{3}}$ we conclude

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\delta(d)\varphi_\epsilon(e))^{\frac{1}{3}}. \quad \text{\textcolor{blue}{\{eq:abd_multiplied\}}} \quad (120)$$

Now let $(\gamma^*, \epsilon^*, \phi^*)$ denote a triple of Young diagrams which maximizes the summand above. Note that the summand is only non-zero if the size constraints $|\alpha| + |\beta| + |\delta| = |\alpha| + |\phi| = |\gamma| + |\delta| = |\epsilon| = k$ are satisfied. As there are at most $(k+1)^{3(n-1)}$ such triples satisfying the size constraints (remember α, β, δ are fixed), we conclude that

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{3(n-1)} \left\| \begin{Bmatrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_\delta(d)\varphi_{\epsilon^*}(e))^{\frac{1}{3}}. \quad \text{\textcolor{blue}{\{eq:first_abd_bound\}}} \quad (121)$$

The second bound is obtained by considering Corollary 58 which states that for any triple (γ, ϵ, ϕ) we have

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) = \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \mathbb{E}_{(X,Y,Z) \sim \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \quad (122)$$

To obtain our desired bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ we make a series of relaxations. First we bound the dimensions of the representations by $\frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \leq \dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n) \leq (k+1)^{3\binom{n}{2}}$. Then we upper bound the average of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over $(X, Y, Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d$ with the maximum of the absolute value. Together this gives

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{3\binom{n}{2}} \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|, \quad \text{\textcolor{blue}{\{eq:max_abs_value_bound\}}} \quad (123)$$

Furthermore, applying Lemma 43 and Proposition 44 to the absolute value $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$ on the right-hand side above produces

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(X)\varphi_\phi(Y+Z)\varphi_\gamma(X+Y)\varphi_\delta(Z)\varphi_\epsilon(X+Y+Z))^{\frac{1}{3}}. \quad \text{\textcolor{blue}{\{eq:sup_bound\}}} \quad (124)$$

Applying Eq. (256) to Eq. (255) gives

$$\begin{aligned} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &\leq (k+1)^{3\binom{n}{2}} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \times \dots \\ &\dots \times \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} (\varphi_\alpha(X)\varphi_\phi(Y+Z)\varphi_\gamma(X+Y)\varphi_\delta(Z)\varphi_\epsilon(X+Y+Z))^{\frac{1}{3}}, \end{aligned} \quad \text{\textcolor{blue}{\{eq:second_abd_bound\}}} \quad (125)$$

Now we choose (γ, ϵ, f) in the above bound to be equal to the optimal choice from $(\gamma^*, \epsilon^*, \phi^*)$ from Eq. (253) and multiply the result with Eq. (253) and then take a square root to obtain

$$\begin{aligned} \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &\leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1)} \times \dots \\ &\dots \times \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} (\varphi_\alpha(a)\varphi_\alpha(X)\varphi_{\phi^*}(f)\varphi_{\phi^*}(Y+Z)\varphi_{\gamma^*}(c)\varphi_{\gamma^*}(X+Y)\varphi_\delta(d)\varphi_\delta(Z)\varphi_{\epsilon^*}(e)\varphi_{\epsilon^*}(X+Y+Z))^{\frac{1}{6}}. \end{aligned} \quad \text{\textcolor{blue}{\{eq:third_abd_bound\}}} \quad (126)$$

Note we have eliminated the $6j$ symbols by using the bound $\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{\infty}^2 \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2$. We are now in a position to apply the eigenvalue separation lemma (specifically Eq. (232)). For each Young diagram $\lambda \in \{\alpha, \delta, \gamma, \epsilon, \phi\}$ we apply Eq. (232) (noting $|\lambda| \leq k$ for each) to conclude

$$\sqrt{\varphi_{\alpha}(a)\varphi_{\alpha}(X)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(a, \text{eig}(X))^{| \alpha |}}{| \alpha |!} \quad (127)$$

$$\sqrt{\varphi_{\delta}(d)\varphi_{\delta}(Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(d, \text{eig}(Z))^{| \delta |}}{| \delta |!} \quad (128)$$

$$\sqrt{\varphi_{\gamma}(c)\varphi_{\gamma}(X+Y)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(c, \text{eig}(X+Y))^{| \gamma |}}{| \gamma |!}, \quad (129)$$

$$\sqrt{\varphi_{\phi}(f)\varphi_{\phi}(Y+Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(f, \text{eig}(Y+Z))^{| \phi |}}{| \phi |!}, \quad (130)$$

$$\sqrt{\varphi_{\epsilon}(e)\varphi_{\epsilon}(X+Y+Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(e, \text{eig}(X+Y+Z))^{| \epsilon |}}{| \epsilon |!}. \quad (131)$$

At this stage, we can combine the bounds for α and ϕ to obtain an expression involving the same power of k :

$$\sqrt{\varphi_{\alpha}(a)\varphi_{\alpha}(X)}\sqrt{\varphi_{\phi}(f)\varphi_{\phi}(Y+Z)} \leq (k+1)^{n(n-1)} \frac{\text{BC}(a, \text{eig}(X))^{| \alpha |}}{| \alpha |!} \frac{\text{BC}(f, \text{eig}(Y+Z))^{| \phi |}}{| \phi |!}, \quad (132)$$

$$\leq (k+1)^{n(n-1)} \frac{[\text{BC}(a, \text{eig}(X)) + \text{BC}(f, \text{eig}(Y+Z))]^k}{k!}, \quad (133)$$

$$= (k+1)^{n(n-1)} \frac{\text{BC}(a \oplus f, \text{eig}(X \oplus Y + Z))^k}{k!}, \quad (134)$$

because $| \alpha | + | \phi | = | \epsilon | = k$. A similar bound also follows from combining γ with δ . Define the maximum product Bhattacharyya coefficient by

$$\text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \text{BC}(a \oplus f, \text{eig}(X \oplus Y + Z)) \text{BC}(c \oplus d, \text{eig}(X + Y \oplus Z)) \text{BC}(e, \text{eig}(X + Y + Z)). \quad (135)$$

From Eq. (258) and the above bounds we obtain

$$\varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d) \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1) + \frac{3}{3}n(n-1)} \frac{1}{k!} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{k}{3}} \quad (136)$$

Now that we have established an upper bound on $\varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d)$ which is uniform in the triple (α, β, δ) , the next step is to sum this expression over all triples (α, β, δ) such that $| \alpha | + | \beta | + | \delta | = k$. Note that summing $\varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d)$ over (α, β, δ) with total size $| \alpha | + | \beta | + | \delta | = k$ yields the analogue of Eq. (250):

$$\sum_{\substack{\alpha, \beta, \delta \\ | \alpha | + | \beta | + | \delta | = k}} \varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d) = \frac{(\text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d])^k}{k!} = \frac{\text{Tr}[e]^k}{k!} \quad (137)$$

where we used that the trace conditions hold, i.e., $\text{Tr}[e] = \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d]$.

As there are at most $(k+1)^{3n}$ triples Young diagrams (α, β, δ) with total size $| \alpha | + | \beta | + | \delta | = k$ being summed here, we finally conclude

$$\frac{\text{Tr}[e]^k}{k!} \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1) + \frac{3}{3}n(n-1) + 3n} \frac{1}{k!} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{k}{3}} \quad (138)$$

or equivalently

$$\frac{1}{\text{Tr}[e]} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{1}{3}} \geq \left(\frac{1}{(k+1)^{\frac{7n^2+11n+6}{4}}} \right)^{\frac{1}{k}}. \quad (139)$$

Finally, using Eq. (84), we can substitute the maximum normalized product of Bhattacharyya coefficients here with minimum normalized sum of distances (as in Eq. (101)) to obtain Eq. (106) which completes the proof. $\square$

5 Inequalities for tetrahedral eigenvalues

`sec:ineqs_from_schur_weyl`

The purpose of this section is to develop the technical machinery used to prove numerous new inequalities constraining the set of tetrahedral eigenvalues. Our approach relies heavily on the theory of irreducible representations of the group $U(n)$ of unitary $n \times n$ matrices, or equivalently, the theory of polynomial irreducible representations of the group $GL(n, \mathbb{C})$ of invertible $n \times n$ matrices with complex entries. More specifically, our approach relies on the decomposition of tensor powers $X^{\otimes k}$ of $n \times n$ matrices provided by Schur-Weyl duality. In Section 5.1 we review this prerequisite representation theory and provide elementary results regarding the decomposition of tensor powers $X^{\otimes k}$ and then afterwards, in Section 6.1, we describe how these decompositions of $X^{\otimes k}$ encode information about the eigenvalues x of X . Building upon this machinery, in Section 5.2, we state and prove a binomial theorem concerning various decompositions of tensors powers of matrix sums $(X + Y)^{\otimes k}$. Using the binomial theorem, in Section 5.3, we will derive new inequalities constraining the set of eigenvalue triples (α, β, γ) of positive semidefinite matrices (A, B, C) satisfying $A + B = C$ relevant for Horn's original eigenvalue problem. Afterwards, in Section 5.5, we again leverage the binomial theorem (or rather its multinomial generalization) to derive new inequalities constraining the set of eigenvalue sextuples (a, b, c, d, e, f) of positive semidefinite matrices (A, B, C, D, E, F) satisfying $A + B = C$, $B + D = F$, $C + D = E$, and $A + F = E$ relevant for the tetrahedral Horn problem.

5.1 Schur-Weyl duality

`sec:schur_weyl`

The purpose of this section is to review the representation theory of $U(n)$, or equivalently, the polynomial representation theory of $GL(n)$. The isomorphism classes of irreducible polynomial representations of $GL(n)$ are indexed by Young diagrams.

A Young diagram λ of size k is a collection of k boxes arranged in rows of sorted length and can be identified with its sequence of non-zero row-lengths, $\lambda \simeq (\lambda_1, \dots, \lambda_\ell)$, where λ_i counts the number of boxes in the i th row, $\lambda_1 \geq \dots \geq \lambda_\ell > 0$, and where $\sum_i \lambda_i = k$. The number of non-zero rows in λ is called the depth or length of λ and is denoted by $\ell(\lambda)$ and the size of λ is denoted by $|\lambda|$. As a shorthand, we write $\lambda \vdash_n k$ to denote a Young diagram λ of size $|\lambda| = k$ with depth at most $\ell(\lambda) \leq n$. For example, $\lambda = \boxplus \simeq (4, 2, 1)$ is a Young diagram of size $|\lambda| = 4 + 2 + 1 = 7$ and depth 3.

For each Young diagram ℓ of size k with depth at most n , there is an irreducible polynomial representation of $U(n)$ (or $GL(n, \mathbb{C})$), denoted by $(\pi_\lambda, V_\lambda^n)$, whose matrix coefficients are polynomials of homogeneous degree k in the entries of $U \in U(n)$. The dimension of the V_λ^n is given by Weyl's dimension formula for $U(n)$ [Hal08; IN66],

$$\dim(V_\lambda^n) = \frac{\Delta(\lambda + \rho)}{\Delta(\rho)} = \frac{\prod_{1 \leq i < j \leq n} (\lambda_i - i - \lambda_j + j)}{\prod_{1 \leq i < j \leq n} (j - i)}, \quad \text{\texttt{\{eq:weyl_dim\}}} \quad (140)$$

where $\rho \simeq (n - 1, n - 2, \dots, 1, 0)$ is known as the Weyl vector and where

$$\Delta(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j). \quad (141)$$

is known as the Vandermonde determinant. The defining representation of $U(n)$, wherein $U \in U(n)$ acts via matrix-vector multiplication on $\mathbb{C}^n$, is associated to the Young diagram with one box, $\lambda = \square \simeq (1)$, and as dimension $\dim(V_\square^n) = n$ as $V_\square^n \simeq \mathbb{C}^n$.

The so-called Schur-Weyl decomposition of $(V_\square^n)^{\otimes k} \simeq (\mathbb{C}^n)^{\otimes k}$ into irreducible representations of $U(n)$ is as follows.

Proposition 27 (Schur-Weyl duality). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible subrepresentations of $S_k \times U(n)$, known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda \vdash_n k} W_\lambda \otimes V_\lambda^n, \quad \text{\texttt{\{eq:schur_weyl_decomp\}}} \quad (142)$$

where the direct summands are indexed by Young diagrams λ of size k and depth at most n rows, and where W_λ supports an irreducible representation of S_k and V_λ^n supports the irreducible representation of $U(n)$.

The dimension of the irreducible representation W_λ of S_k is given by

$$\dim(W_\lambda) = \frac{|\lambda|!}{H_\lambda}, \quad H_\lambda = \prod_{b \in \lambda} h_\lambda(b), \quad \text{\texttt{seq:hook_dim}} \quad (143)$$

where the hook length, $h_\lambda(b)$, of b counts the number of boxes to the right and below b plus one for itself. We refer to H_λ as the hook-product for λ .

Now let $(\pi_\lambda, V_\lambda^n)$ be an irreducible representation of $U(n)$ labelled by the Young diagram λ with depth at most n . Throughout this paper, the representation spaces V_λ^n are assumed to be equipped with respective Hermitian inner products making π_λ *unitary* representations of $U(n)$. Similarly, if $(\sigma_\lambda, W_\lambda)$ denotes the irreducible representation of S_k labelled by λ , we assume W_λ is equipped with a real inner product making σ_λ an *orthogonal* representation of S_k . In particular, the defining representation $V_\square \simeq \mathbb{C}^n$ of $U(n)$ is equipped with the standard Hermitian inner product on $\mathbb{C}^n$ making the standard basis $\{e_1, \dots, e_n\}$ orthonormal. Then for each λ -isotypic subspace of $V_\square^{\otimes |\lambda|} \simeq (\mathbb{C}^n)^{\otimes |\lambda|}$, we let

$$\iota_\lambda : W_\lambda \otimes V_\lambda^n \rightarrow V_\square^{\otimes |\lambda|}, \quad (144)$$

denote the unique $S_k \times U(n)$ -equivariant map from $W_\lambda \otimes V_\lambda^n$ to $V_\square^{\otimes |\lambda|}$ which is additionally an isometry with respect to the chosen inner products on W_λ , V_λ^n and $V_\square$.

Then given an $n \times n$ matrix X , viewed as a linear operator on the defining representation $V_\square^n \simeq \mathbb{C}^n$, let $\pi_\lambda(X)$ be defined by

$$\pi_\lambda(X) := \frac{\text{Tr}_{W_\lambda}[\iota_\lambda^\dagger X^{\otimes |\lambda|} \iota_\lambda]}{\dim(W_\lambda)}.$$

Note that when X is invertible, i.e. $X \in \text{GL}(n, \mathbb{C})$, the operator $\pi_\lambda(X)$ as defined above coincides with the representation of X on V_λ^n , so this abuse of notation is permitted. For any positive integer k , the k -th tensor power of X then admits a decomposition as $X^{\otimes k} \simeq \bigoplus_{\lambda \vdash_n k} \pi_\lambda(X)^{\oplus W_\lambda}$, or more precisely,

$$X^{\otimes k} = \sum_{\lambda \vdash_n k} \iota_\lambda(\pi_\lambda(X) \otimes \text{id}_{W_\lambda}) \iota_\lambda^\dagger. \quad (145)$$

As a special case, when $\lambda = \square$, then $\pi_\square(X) = X$. Note also that the matrix coefficients of $\pi_\lambda(X)$ are homogeneous degree $|\lambda|$ polynomials in the matrix coefficients of X meaning for any scalar $s \in \mathbb{C}$,

$$\pi_\lambda(sX) = s^{|\lambda|} \pi_\lambda(X). \quad (146)$$

Additionally, if X, Y are two matrices, we have

$$\pi_\lambda(XY) = \pi_\lambda(X) \pi_\lambda(Y). \quad (147)$$

The corresponding character $\chi_\lambda(X) = \text{Tr}(\pi_\lambda(X))$ of a representation of a Hermitian matrix X with eigenvalues x is equal to

$$\chi_\lambda(X) := \text{Tr}(\pi_\lambda(X)) = \frac{\text{Tr}[P_\lambda X^{\otimes |\lambda|}]}{\dim(W_\lambda)} = s_\lambda(x),$$

where $P_\lambda := \iota_\lambda^\dagger \iota_\lambda$ is the orthogonal projection operator onto the λ -isotypic subspace, and where $s_\lambda(x) = s_\lambda(x_1, \dots, x_n)$ is known as the Schur-polynomial associated to λ .

5.2 The binomial theorem

\text{\texttt{sec:binomial_theorem}}

The purpose of this section is to state and prove the binomial theorem, which serves as the foundation for all subsequent results in this paper. The binomial theorem expresses any polynomial irreducible representation of a matrix sum $X + Y$ as a sum over the polynomial irreducible representations of the summands X and Y .

The theorem inherits its name from the non-commutative binomial expansion of the k -th tensor power of the sum $X + Y$,

$$(X + Y)^{\otimes k} = \sum_{\ell=0}^k \binom{k}{\ell} \left[\frac{1}{k!} \sum_{\sigma \in S_k} T(\sigma)(X^{\otimes \ell} \otimes Y^{\otimes (k-\ell)})T(\sigma^{-1}) \right] \quad (148)$$

where $T(\sigma)$ denotes the operator which acts on $(\mathbb{C}^n)^{\otimes k}$ by a σ -permutation of the k tensor factors of $\mathbb{C}^n$. For an example, consider the case where $k = 3$:

$$\begin{aligned} (X + Y)^{\otimes 3} = & \underbrace{X \otimes X \otimes X}_{\ell=3} + \underbrace{X \otimes X \otimes Y + X \otimes Y \otimes X + Y \otimes X \otimes X}_{\ell=2} + \\ & + \underbrace{X \otimes Y \otimes Y + Y \otimes X \otimes Y + Y \otimes Y \otimes X}_{\ell=1} + \underbrace{Y \otimes Y \otimes Y}_{\ell=0}. \end{aligned} \quad (149)$$

As we have seen previously, the k -th tensor power of a matrix can also be decomposed into a sum over polynomial irreducible representations labelled by Young diagrams γ with sizes $|\gamma| = k$. Combining this formula with the non-commutative binomial expansion yields

$$(X + Y)^{\otimes k} = \sum_{\gamma \vdash k} \iota_\gamma(\pi_\gamma(X + Y) \otimes \text{id}_{W_\gamma}) \iota_\gamma^\dagger \quad (150)$$

where

$$\dim(W_\gamma) \pi_\gamma(X + Y) = \sum_{\ell=0}^k \binom{k}{\ell} \text{Tr}_{W_\gamma} [\iota_\gamma^\dagger (X^{\otimes \ell} \otimes Y^{\otimes (k-\ell)}) \iota_\gamma], \quad (151)$$

where we have eliminated the symmetrization operation over $\sigma \in S_k$ because the partial trace over W_γ is S_k -invariant. At this stage, for each summand indexed by $0 \leq \ell \leq k$, we can decompose $X^{\otimes \ell}$ into representations of X labelled by α with sizes $|\alpha| = \ell$ and $Y^{\otimes \ell}$ into representations of Y labelled by β with sizes $|\beta| = k - \ell$.

$$X^{\otimes \ell} = \sum_{\alpha \vdash \ell} \iota_\alpha(\pi_\alpha(X) \otimes \text{id}_{W_\alpha}) \iota_\alpha^\dagger, \quad Y^{\otimes k-\ell} = \sum_{\beta \vdash k-\ell} \iota_\beta(\pi_\beta(Y) \otimes \text{id}_{W_\beta}) \iota_\beta^\dagger. \quad (152)$$

Upon doing so, one encounters the composite operator $\iota_\gamma^\dagger(\iota_\alpha \otimes \iota_\beta)$ which maps from $(V_\alpha^n \otimes W_\alpha) \otimes (V_\beta^n \otimes W_\beta)$ to $(V_\gamma^n \otimes W_\gamma)$ in an $S_\ell \times S_{k-\ell}$ -equivariant and $U(n)$ -equivariant manner. At this stage define multiplicity spaces $C_{\alpha\beta}^\gamma$ and $\tilde{C}_{\alpha\beta}^\gamma$ by

$$C_{\alpha\beta}^\gamma := \text{Hom}_{U(n)}(V_\gamma^n, V_\alpha^n \otimes V_\beta^n), \quad (153)$$

$$\tilde{C}_{\alpha\beta}^\gamma := \text{Hom}_{S_\ell \times S_{k-\ell}}(W_\alpha \otimes W_\beta, W_\gamma), \quad (154)$$

where $C_{\alpha\beta}^\gamma$ and $\tilde{C}_{\alpha\beta}^\gamma$ are equipped with Hermitian inner products induced by the corresponding inner products on each of the representation spaces. Furthermore, let $\iota_{\alpha\beta}^\gamma : C_{\alpha\beta}^\gamma \otimes V_\gamma^n \rightarrow V_\alpha^n \otimes V_\beta^n$ be the $U(n)$ -equivariant isometry for the γ -isotypic subspace of $V_\alpha^n \otimes V_\beta^n$ and $\omega_{\alpha\beta}^\gamma : C_{\alpha\beta}^\gamma \otimes W_\alpha \otimes W_\beta \rightarrow W_\gamma$ the $S_\ell \times S_{k-\ell}$ -equivariant isometry for the (α, β) -isotypic subspace of W_γ .

An important consequence of Schur-Weyl duality is an identification between $C_{\alpha\beta}^\gamma$ and $\tilde{C}_{\alpha\beta}^\gamma$ ⁵ given by the map $T_{\alpha\beta}^\gamma : C_{\alpha\beta}^\gamma \rightarrow \tilde{C}_{\alpha\beta}^\gamma$, where

$$T_{\alpha\beta}^\gamma := \frac{\text{Tr}_{W_\alpha \otimes W_\beta \otimes V_\gamma^n} \left[(\omega_{\alpha\beta}^\gamma)^\dagger \otimes \text{id}_{V_\gamma^n} \right] \circ (\iota_\gamma^\dagger) \circ (\iota_\alpha \otimes \iota_\beta) \circ (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{W_\alpha \otimes W_\beta})}{\dim(W_\alpha \otimes W_\beta \otimes V_\gamma^n)}. \quad (155)$$

Using Schur's lemma, one can show that $T_{\alpha\beta}^\gamma$ is actually an isometric and bijective map from $C_{\alpha\beta}^\gamma$ and $\tilde{C}_{\alpha\beta}^\gamma$. Combining everything together, we obtain

$$\frac{\dim W_\gamma \pi_\gamma(X + Y)}{|\gamma|!} = \sum_{\alpha, \beta} \text{Tr}_{C_{\alpha, \beta}^\gamma} \left[(\iota_{\alpha, \beta}^\gamma)^\dagger \frac{\dim W_\alpha \pi_\alpha(X)}{|\alpha|!} \otimes \frac{\dim W_\beta \pi_\beta(Y)}{|\beta|!} (\iota_{\alpha, \beta}^\gamma) \right], \quad (156)$$

⁵At least whenever $|\gamma| = k$, $|\alpha| = \ell$ and $|\gamma| = k - \ell$ and $\ell(\alpha), \ell(\beta), \ell(\gamma) \leq n$.

where we leave the summation over all pairs (α, β) such that $|\alpha| = \ell$, $|\beta| = k - \ell$ for $|\gamma| = k$ implicit simply because the multiplicity space $C_{\alpha\beta}^\gamma$ is empty if $|\alpha| + |\beta| \neq |\gamma|$ and thus the corresponding summand is taken to be zero.

In the remaining of the paper, we will use the above formula, Eq. (156), to prove numerous results concerning the eigenvalues of Hermitian matrices and their sums. In order to simplify the presentation of our results considerably, we introduce the following definitions.

Definition 28 (Hook-rescaled representations). Given an $n \times n$ matrix X and a Young diagram λ , define an operator $\Phi_\lambda(X)$ acting on the representation space V_λ^n as:

$$\Phi_\lambda(X) := \frac{\pi_\lambda(X)}{H_\lambda}, \quad (157)$$

where $H_\lambda = \prod_{b \in \lambda} h_\lambda(b)$ is the product of the Hook-lengths of the boxes in the Young diagram λ . Furthermore, the trace of $\Phi_\lambda(X)$ be

$$\varphi_\lambda(X) = \text{Tr}[\Phi_\lambda(X)] = \frac{s_\lambda(x)}{H_\lambda}. \quad (158)$$

Remark 29. The operator $\Phi_\lambda(X)$ and trace $\varphi_\lambda(X)$ are simply rescalings of the representation matrix $\pi_\lambda(X)$ of X and the character $\chi_\lambda(X)$ of X respectively. This normalization scheme is such that summing the rescaled character over all Young diagrams λ with fixed size k or over all sizes yields respectively

$$\sum_{\lambda \vdash k} \varphi_\lambda(X) = \frac{\text{Tr}(X)^k}{k!}, \quad \sum_{\lambda} \varphi_\lambda(X) = \exp(\text{Tr}(X)). \quad (159)$$

Before stating the main result of this section, we first generalize the quantities $\Phi_\lambda(X)$ and $\varphi_\lambda(X)$ to consider arguments involving more than one matrix.

Definition 30. Given a pair (X, Y) of $n \times n$ matrices and a triple of Young diagrams (α, β, γ) , define an operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ acting on the representation space V_γ^n as

$$\Phi_\gamma^{\alpha\beta}(X, Y) := \text{Tr}_{C_{\alpha,\beta}^\gamma} [\iota_{\alpha,\beta}^{\gamma;\dagger}(\Phi_\alpha(X) \otimes \Phi_\beta(Y)) \iota_{\alpha,\beta}^\gamma], \quad (160)$$

where $\iota_{\alpha,\beta}^\gamma : C_{\alpha,\beta}^\gamma \otimes V_\alpha^n \otimes V_\beta^n \rightarrow V_\gamma^n \otimes V_\alpha^n \otimes V_\beta^n$ is the $U(n)$ -equivariant isometry for the γ -isotypic subspace of $V_\alpha^n \otimes V_\beta^n$. Note the sum over Young diagrams α , and β is implicitly restricted to those satisfying $|\alpha| + |\beta| = |\gamma|$, since otherwise the multiplicity space $C_{\alpha,\beta}^\gamma$ is zero-dimensional and thus the summand is taken to be zero. Analogously, let the trace of $\Phi_\gamma^{\alpha\beta}(X, Y)$ be denoted by

$$\varphi_\gamma^{\alpha\beta}(X, Y) := \text{Tr}[\Phi_\gamma^{\alpha\beta}(X, Y)] = \text{Tr}[P_{\alpha,\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))], \quad (161)$$

where $P_{\alpha,\beta}^\gamma = \iota_{\alpha,\beta}^\gamma \iota_{\alpha,\beta}^{\gamma;\dagger}$ is the projection operator onto the γ -isotypic subspace of $V_\alpha^n \otimes V_\beta^n$.

Remark 31. While $\Phi_\lambda(X)$ is a homogeneous polynomial in the matrix coefficients of X of degree $|\lambda|$ in the sense that $\Phi_\lambda(sX) = s^{|\lambda|} \Phi_\lambda(X)$ holds for any $s \in \mathbb{R}$, $\Phi_\gamma^{\alpha\beta}(X, Y)$ is a bihomogeneous polynomial in X and Y of degrees $|\alpha|$ and $|\beta|$ respectively in the sense that for any $s, t \in \mathbb{R}$,

$$\Phi_\gamma^{\alpha\beta}(sX, tY) = s^{|\alpha|} t^{|\beta|} \Phi_\gamma^{\alpha\beta}(X, Y). \quad (162)$$

rem:trivial_yd

Remark 32. The quantity $\Phi_\gamma^{\alpha\beta}(X, Y)$ generalizes $\Phi_\lambda(X)$ in the sense that one can take the triple $(\alpha, \beta, \gamma) = (\lambda, (), \lambda)$ (where $()$ denotes the Young diagram with zero boxes), in which case $V_{()}^n$ supports the trivial one-dimensional representation with $\Phi_{()}(Y) = 1$ and because $\dim(C_{\lambda,()}^\lambda) = 1$,

$$\forall Y : \quad \Phi_\lambda^{\lambda,()}(X, Y) = \Phi_\lambda(X). \quad (163)$$

With these definitions in place, we are in a position to state the binomial theorem.

Theorem 33 (Binomial theorem). *For all $n \times n$ matrices X, Y , and all Young diagrams γ ,*

$$\Phi_\gamma(X + Y) = \sum_{\alpha, \beta} \Phi_\gamma^{\alpha\beta}(X, Y), \quad (164)$$

where the sum is implicitly taken over all Young diagrams such that $|\alpha| + |\beta| = |\gamma|$.

Proof. See preceding discussion. $\square$

Taking the trace of the operators in Theorem 33 produces the following corollary which pertains to sums of the quantity $\varphi_\gamma^{\alpha\beta}(X, Y)$ (Eq. (161)).

cor:bisummation

Corollary 34. *Let X, Y be $n \times n$ matrices. Then the following identities hold:*

$$\forall \alpha, \beta : \quad \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y) = \varphi_\alpha(X) \varphi_\beta(Y), \quad (165)$$

$$\forall \gamma : \quad \sum_{\alpha, \beta} \varphi_\gamma^{\alpha\beta}(X, Y) = \varphi_\gamma(X + Y). \quad (166)$$

Proof. The first identity involving a summation over α and β is simply the result of taking the trace of both sides of Theorem 33. For the second identity, involving a summation over γ , follows from the resolution of the identity $\sum_{\gamma \vdash |\alpha| + |\beta|} P_{\alpha\beta}^\gamma = \text{id}_{V_\alpha^n \otimes V_\beta^n}$:

$$\varphi_\alpha(X) \varphi_\beta(Y) = \text{Tr}[\Phi_\alpha(X) \otimes \Phi_\beta(Y)] = \sum_{\gamma} \text{Tr}[P_{\alpha\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))] = \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y), \quad (167)$$

where the last equality is just the definition of $\varphi_\gamma^{\alpha\beta}(X, Y)$. Once again, the condition that $|\gamma| = |\alpha| + |\beta|$ is implicitly present. $\square$

Remark 35. One might consider what happens to the operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ when one sums over γ instead of (α, β) . However, an analogous identity concerning a sum over γ of the operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ does not immediately follow because the tensor product $\Phi_\alpha(X) \otimes \Phi_\beta(Y)$ is not necessarily block diagonalizable into γ -isotypic subspaces in any obvious way. One exception occurs when $X = Y$, in which case,

$$\pi_\alpha(X) \otimes \pi_\beta(X) = \sum_{\gamma} \iota_{\alpha\beta}^\gamma(\pi_\gamma(X) \otimes \text{id}_{C_{\alpha\beta}^\gamma}) \iota_{\alpha\beta}^{\gamma\dagger}, \quad (168)$$

which is the operator version of the character product formula $\chi_\alpha(X) \chi_\beta(X) = \sum_{\gamma} c_{\alpha\beta}^\gamma \chi_\gamma(X)$.

5.3 Inequalities for Horn's original problem

sec:new_ineq_horn_orig

In Section 5.2 we introduced to the quantity $\varphi_\gamma^{\alpha\beta}(X, Y)$ (Eq. (161)) and saw in Corollary 34 that it satisfied two summation identities. Under the assumption that the matrices X , and Y are positive semidefinite, we can conclude that $\varphi_\gamma^{\alpha\beta}(X, Y)$ is non-negative also because $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))]$ where $P_{\alpha\beta}^\gamma$ is a projection operator and $\Phi_\alpha(X), \Phi_\beta(Y) \geq 0$ because $X, Y \geq 0$. Moreover, we note that $\varphi_\gamma^{\alpha\beta}(X, Y) = 0$ whenever the Littlewood-Richardson coefficient $c_{\alpha\beta}^\gamma$ vanishes. These observations imply the following necessary condition for compatibility of eigenvalues.

prop:coupling_condition

Proposition 36. *Let (a, b, c) be a triple of non-negative eigenvalues. Then there exists positive semidefinite $n \times n$ matrices (A, B, C) satisfying $A + B = C$ with eigenvalues (a, b, c) only if for every triple of Young diagrams (α, β, γ) there exists a non-negative $\Gamma_{\alpha\beta}^\gamma \geq 0$ such that $\Gamma_{\alpha\beta}^\gamma = 0$ whenever $c_{\alpha\beta}^\gamma = 0$ and*

$$\forall \alpha, \beta : \quad \sum_{\gamma} \Gamma_{\alpha\beta}^\gamma = \varphi_\alpha(a) \varphi_\beta(b), \quad \text{and} \quad \forall \gamma : \quad \sum_{\alpha, \beta} \Gamma_{\alpha\beta}^\gamma = \varphi_\gamma(c). \quad (169)$$

Proof. The claim follows directly from Corollary 34 after identifying $\Gamma_{\alpha\beta}^\gamma = \varphi_\gamma^{\alpha\beta}(A, B) \geq 0$. $\square$

Corollary 37. Let (a, b, c) be the eigenvalues of positive semidefinite $n \times n$ matrices (A, B, C) satisfying $A + B = C$. Then,

$$\forall \alpha, \beta: \quad \varphi_\alpha(a)\varphi_\beta(b) \leq \sum_{\substack{\gamma \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\gamma(c), \quad \text{and} \quad \forall \gamma: \quad \varphi_\gamma(c) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(a)\varphi_\beta(b). \quad (170)$$

Proof. The claim follows from Proposition 36 and noting that whenever $\Gamma_{\alpha\beta}^\gamma > 0$ (only when $c_{\alpha\beta}^\gamma > 0$) we have the bound $\Gamma_{\alpha\beta}^\gamma \leq \min\{\varphi_\alpha(a)\varphi_\beta(b), \varphi_\gamma(c)\}$ for all triples (α, β, γ) . $\square$

Remark 38. Using Proposition 36, it is possible to prove even stronger inequalities than those appearing in Corollary 37 by examining the set of triples of Young diagrams (α, β, γ) where $c_{\alpha, \beta}^\gamma \neq 0$. We illustrate this point with an example. Let $k = |\alpha| + |\beta| = |\gamma| = 4$, then the following is the table of values for $c_{\alpha\beta}^\gamma$.

$c_{\alpha\beta}^\gamma$	α	β	γ

Using this table, one sees the inequalities in Corollary 37 are associated to single columns or single rows, e.g., along the $(\alpha, \beta) = (\square, \square)$, we obtain

$$\varphi_{\square}(X)\varphi_{\square}(Y) \leq \varphi_{\square\square}(X+Y) + \varphi_{\square\square}(X+Y). \quad (171)$$

However, using Proposition 36, a stronger inequality is obtained by including all columns covered by the rows associated to the right-hand side of the above inequality, e.g.,

$$\varphi_{\square\square}(X) + \varphi_{\square\square}(X) + \varphi_{\square}(X)\varphi_{\square}(Y) + \varphi_{\square}(X)\varphi_{\square}(Y) + \varphi_{\square\square}(Y) + \varphi_{\square\square}(Y) \leq \varphi_{\square\square}(X+Y) + \varphi_{\square\square}(X+Y). \quad (172)$$

In general, Proposition 36 describes a linear system of inequality constraints in the variables $\Gamma_{\alpha\beta}^\gamma, \varphi_\alpha(a)\varphi_\beta(b)$, and $\varphi_\gamma(c)$ which, in principle, be solved for $\varphi_\alpha(a)\varphi_\beta(b)$ and $\varphi_\gamma(c)$ by standard linear quantifier elimination techniques such as Fourier-Motzkin elimination [DE+72]. While these additional inequalities are certainly interesting to explore, for the purposes of the original Horn problem, we will later prove, in Proposition 59, that it is sufficient to restrict our attention to inequalities of the form Corollary 37.

5.4 The $6j$ symbol for $U(n)$

sec:sixj_tet

The purpose of this section is to introduce a representation theoretic object, namely the $6j$ symbol, and then in the subsequent sections show how it places constraints on whether a given sextuple of eigenvalues (a, b, c, d, e, f) is tetrahedral in a manner which generalizes how the Littlewood-Richardson coefficient encodes whether a given triple (a, b, c) is trigonal. For the case where $n = 2$, this correspondence is well-known [Rob99; PR68; Bac10], while for the case where $n \geq 3$, this correspondence was already established indirectly by Christandl, Şahinoğlu, and Walter [CŞW18]. Here we establish this correspondence directly.

As we discussed in the introduction, the $6j$ symbol emerges naturally when considering successive decompositions of the triple tensor product $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$.

defn:sixj

Definition 39. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams with sizes satisfying

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\gamma| + |\delta| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad (173)$$

The $U(n)$ -6j operator is the operator

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} : C_{\alpha,\beta}^\gamma \otimes C_{\delta,\gamma}^\epsilon \rightarrow C_{\alpha,\phi}^\epsilon \otimes C_{\beta,\delta}^\phi, \quad (174)$$

defined by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{1}{\dim(V_\epsilon)} \text{Tr}_{V_\epsilon}[(\iota_{\alpha\phi}^\epsilon \otimes \text{id}_{C_{\beta\delta}^\phi})^\dagger (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi)^\dagger (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) (\text{id}_{C_{\alpha\beta}^\gamma} \otimes \iota_{\gamma\delta}^\epsilon)]. \quad \text{\texttt{eq:sixj}} \quad (175)$$

To illustrate the situation more concretely, consider the following three projection operators on the triple tensor product $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$:

$$P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}, \quad P_{\alpha\beta\delta}^\epsilon, \quad \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi. \quad (176)$$

In general, $P_{\alpha\beta}^\gamma$ and $P_{\beta\delta}^\phi$ do not necessarily commute with each other in the sense that for some values of $\alpha, \beta, \delta, \gamma$, and ϕ , one observes

$$[P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}, \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi] \neq 0. \quad (177)$$

Nevertheless, the projection operators $P_{\alpha\beta}^\gamma$ and $P_{\beta\delta}^\phi$ each respectively commute with $P_{\alpha\beta\delta}^\epsilon$ in the sense that for all $\alpha, \beta, \gamma, \delta, \epsilon$, and ϕ ,

$$[P_{\alpha\beta\delta}^\epsilon, P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}] = 0, \quad [P_{\alpha\beta\delta}^\epsilon, \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi] = 0. \quad (178)$$

Let $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ and $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$ be Hermitian projection operators $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$ given by

$$\begin{aligned} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} &:= P_{\alpha\beta\delta}^\epsilon (P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) = (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) (P_{\gamma\delta}^\epsilon \otimes \text{id}_{C_{\alpha\beta}^\gamma}) (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n})^\dagger, \\ \Pi_{\alpha\beta\delta}^{\epsilon;\phi} &:= P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi) = (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi) (P_{\alpha\phi}^\epsilon \otimes \text{id}_{C_{\beta\delta}^\phi}) (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi)^\dagger, \end{aligned} \quad \text{\texttt{eq:double_projs}} \quad (179)$$

For any operator $O : V_1 \rightarrow V_2$ between Hermitian vector spaces V_1 and V_2 , we define operator norms $\|O\|_\infty = \max_{v \in V_1, \|v\|_{V_1}=1} \|Ov\|_{V_2}$ and $\|O\|_2 = \text{Tr}_1[O^\dagger O]^\frac{1}{2} = \text{Tr}_2[OO^\dagger]^\frac{1}{2}$.

prop:6j_norm_double_proj

Proposition 40. For any sextuple of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$,

$$\left\| \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \right\|_\infty = \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_\infty, \quad \left\| \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \right\|_2^2 = \frac{1}{\dim V_\epsilon} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}]. \quad \text{\texttt{eq:6j_norm_equivalences}} \quad (180)$$

Proof. By definition, the product $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}$ is an $U(n)$ -equivariant map on $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$ supported on the ϵ -isotypic subspace $V_\epsilon^n \otimes C_{\alpha\beta\delta}^\epsilon$. By Schur's lemma and Definition 39, we conclude

$$\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} = (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi) (\iota_{\alpha\phi}^\epsilon \otimes \text{id}_{C_{\beta\delta}^\phi}) \left(\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \otimes V_\epsilon^n \right) (\text{id}_{C_{\alpha\beta}^\gamma} \otimes \iota_{\gamma\delta}^\epsilon)^\dagger (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n})^\dagger. \quad (181)$$

The norm conditions in Eq. (180) then follow because the norms $\|O\|_\infty$ and $\|O\|_2$ are invariant under embedding O into larger spaces using isometries and moreover that $\|O \otimes \text{id}_V\|_\infty = \|O\|_\infty$ for any vector space V . $\square$

5.5 Inequalities for the tetrahedral Horn problem

sec:new_ineq_horn_tet

In Section 5.3, we saw how the quantity $\varphi_\gamma^{\alpha\beta}(X, Y)$ (Eq. (161)) could be used to produce inequalities applicable to the original Horn problem. In this section, we generalize this analysis to produce inequalities applicable to the tetrahedral Horn problem using a generalization of $\varphi_\gamma^{\alpha\beta}(X, Y)$ which depends on three matrices.

Definition 41. Let X, Y, Z be $n \times n$ matrices. For any sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of Young diagrams, let

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \text{Tr}[(P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi) \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)]. \quad (182)$$

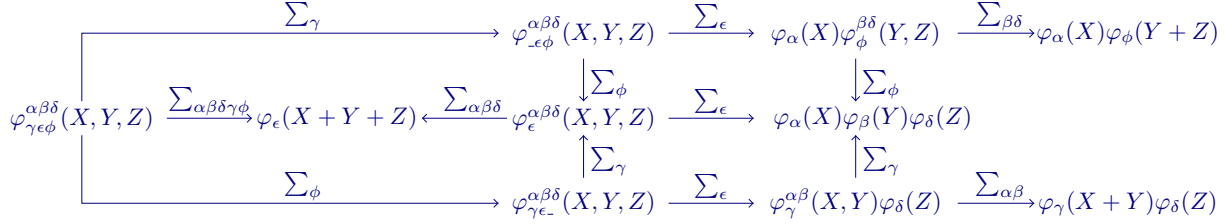


Figure 3: Various summation identities satisfied by $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$.

`fig:sum_ident`

Remark that in our definition of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we permit the sizes of the involved Young diagrams to be arbitrary. However, there are numerous conditions on the sizes of the Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are necessary (but not always sufficient) for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ to be non-zero; specifically,

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\phi| = |\epsilon|, \quad |\gamma| + |\delta| = |\epsilon|. \quad (183)$$

`{eq:yd_sizes_tet}`

If we additionally let

$$\Omega_{\alpha\beta\delta}^{X,Y,Z} := \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z). \quad (184)$$

then we can compactly express $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ as

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]. \quad (185)$$

`{eq:varphi_compact_tet}`

Unlike the quantity $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma \Phi_\alpha(X) \otimes \Phi_\beta(Y)]$ involving two matrices X, Y (see Section 5.2), the quantity defined here, $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, is not necessarily non-negative or even real valued even under the assumption that (X, Y, Z) are positive semidefinite. Nevertheless, summing over either γ or ϕ restores non-negativity because it eliminates one of the potentially non-commuting projection operators $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ or $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. We denote these partial sums of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over γ and ϕ respectively as

$$\begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &:= \sum_\gamma \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\gamma\delta}^\epsilon (\Phi_\gamma^{\alpha\beta}(X, Y) \otimes \Phi_\delta(Z))], \\ \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) &:= \sum_\phi \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\alpha\phi}^\epsilon (\Phi_\alpha(X) \otimes \Phi_\phi^{\beta\delta}(Y, Z))], \end{aligned} \quad (186)$$

`{eq:psum_defns}`

where the sum over γ is implicitly such that $|\gamma| = |\alpha| + |\beta|$ and the sum over ϕ is implicitly such that $|\phi| = |\beta| + |\delta|$. Although $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ is not necessarily real or positive, the above expression makes it evident that

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \geq 0, \quad \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \geq 0. \quad (187)$$

The various summation identities satisfied by Fig. 3 are summarized in Fig. 3.

The next series of results concerns upper bounds on the absolute value of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$. These upper bounds follow from the following general lemma applied to the compact expression for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ appearing in Eq. (185).

`prop:cauchy_schwarz_ineq_double_proj`

Proposition 42. *Let $\Omega \geq 0$ be a positive semidefinite operator and let Π_1, Π_2 be Hermitian projection operators acting on a common vector space. Then*

$$|\text{Tr}(\Pi_1 \Pi_2 \Omega)| \leq \|\Pi_1 \Pi_2\|_\infty \text{Tr}(\Pi_1 \Omega)^{\frac{1}{2}} \text{Tr}(\Pi_2 \Omega)^{\frac{1}{2}}. \quad (188)$$

Proof. As $\Omega \geq 0$ is positive semidefinite, let $\Omega^{\frac{1}{2}} \geq 0$ denote its square root. Then let $L = \Omega^{\frac{1}{2}} \Pi_2 \Pi_1$ and $R = \Omega^{\frac{1}{2}} \Pi_1$ so that

$$\langle L, R \rangle = \text{Tr}(L^\dagger R) = \text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_1) = \text{Tr}(\Pi_1 \Pi_2 \Omega), \quad (189)$$

where the last equality uses the assumption that $\Pi_1^2 = \Pi_1$. An application of the Cauchy-Schwarz inequality for the Hilbert-Schmidt inner product $\langle L, R \rangle$ then implies

$$|\text{Tr}(\Pi_1 \Pi_2 \Omega)|^2 = |\langle L, R \rangle|^2 \leq \text{Tr}[L^\dagger L] \text{Tr}[R^\dagger R] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1] \text{Tr}[\Pi_1 \Omega \Pi_1] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] \text{Tr}[\Pi_1 \Omega]. \quad (190)$$

Now let $\{v_j\}_j$ be an orthonormal eigenbasis for $\Pi_2 \Omega \Pi_2$ with eigenvalue $\omega_j \geq 0$. Then, because it was assumed that $\Pi_2^2 = \Pi_2$, we have

$$\text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] = \text{Tr}[(\Pi_2 \Pi_1^2 \Pi_2)(\Pi_2 \Omega \Pi_2)] = \sum_j \omega_j \langle \Pi_1 \Pi_2 v_j, \Pi_1 \Pi_2 v_j \rangle \leq \sum_j \omega_j \|\Pi_1 \Pi_2\|_\infty^2 = \text{Tr}[\Pi_2 \Omega] \|\Pi_1 \Pi_2\|_\infty^2. \quad (191)$$

The claim then follows from combining these inequalities and taking the square root of both sides. $\square$
lem:varphi_tet_main_ineq

Lemma 43. *Let (X, Y, Z) be positive semidefinite $n \times n$ matrices. Then for all sextuples of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$,*

$$|\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)| \leq \left\| \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\|_\infty \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}}. \quad (192)$$

Proof. Under the assumption that $X, Y, Z \geq 0$, we have that $\Omega_{\alpha\beta\delta}^{X,Y,Z} = \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z) \geq 0$ is positive semidefinite also. Now applying Proposition 42 to the compact expression for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (185) yields

$$\left| \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] \right| \leq \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_\infty \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \quad (193)$$

which is equivalent to the claim due to Proposition 40. $\square$

Our next proposition enables us to place bounds on $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ which depend only the eigenvalues of $X, Y, X + Y, Z, X + Y + Z$, and $Y + Z$ and not their eigenvectors. *prop:tet_psum_ineq*

Proposition 44. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then we have bounds on the partial sums,*

$$\begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &\leq \varphi_\alpha(X) \varphi_\phi(Y + Z), \\ \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) &\leq \varphi_\gamma(X + Y) \varphi_\delta(Z), \end{aligned} \quad \begin{matrix} \{eq:psum_prod_bounds\} \\ (194) \end{matrix}$$

as well as bounds on $\varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z)$:

$$\max\{\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z), \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)\} \leq \varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z) \leq \min\{\varphi_\epsilon(X + Y + Z), \varphi_\alpha(X) \varphi_\beta(Y) \varphi_\delta(Z)\}. \quad \begin{matrix} \{eq:psum_yc_ye_uniform\} \\ (195) \end{matrix}$$

Proof. Each of these bounds are consequences of the summation identities for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and the fact that the partial sums $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ are non-negative real-valued whenever $X, Y, Z \geq 0$ are positive semidefinite. These summation identities are summarized in Fig. 3. To begin, the quantity $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ (defined in Eq. (186)) can be summed over ϵ to obtain:

$$\sum_\epsilon \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \sum_\epsilon \text{Tr}[P_{\alpha\phi}^\epsilon(\Phi_\alpha(X) \otimes \Phi_\phi^{\beta\delta}(Y, Z))] = \text{Tr}[\Phi_\alpha(X) \otimes \Phi_\phi^{\beta\delta}(Y, Z)] = \varphi_\alpha(X) \varphi_\phi^{\beta\delta}(Y, Z). \quad \begin{matrix} \{eq:psum_yf_over_ye\} \\ (196) \end{matrix}$$

Further summing over β and δ and applying Corollary 34 produces

$$\sum_{\epsilon\beta\delta} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \varphi_\alpha(X) \sum_{\beta\delta} \varphi_\phi^{\beta\delta}(Y, Z) = \varphi_\alpha(X) \varphi_\phi(Y + Z). \quad (197)$$

The other summation identities are analogous. Now using the non-negativity of $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we obtain the first claimed inequality in Eq. (194). The second claimed inequality in Eq. (194) follows analogously from summing $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ over ϵ, α and β . The remaining claimed inequalities in Eq. (195) are the result

of similar summation identities. For example, summing the quantity $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over ϕ and applying Eq. (179) produces

$$\begin{aligned} \sum_{\phi} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}^{\beta\delta}(Y, Z))], \\ &= \sum_{\phi} \text{Tr}[P_{\alpha\beta\delta}^{\epsilon}(\text{id}_{\alpha} \otimes P_{\beta\delta}^{\phi})(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z))], \quad \text{\texttt{feq:psum_yf_over_yf}} \\ &= \text{Tr}[P_{\alpha\beta\delta}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z))], \\ &= \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z). \end{aligned} \quad (198)$$

Therefore, from the non-negativity of $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we obtain $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ which establishes one half of the lower bound on $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (195); the other half of the lower bound follows analogously from summing $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ over γ . Finally, for the upper bounds on $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (195), we remark that Corollary 34 can be extended to handle more than two matrices by repeated application of Corollary 34; specifically, by starting with Eq. (196) and further summing over ϕ and applying Eq. (198) and Corollary 34, we obtain

$$\begin{aligned} \sum_{\epsilon} \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\epsilon\phi} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \\ &= \sum_{\phi} \varphi_{\alpha}(X) \varphi_{\phi}^{\beta\delta}(Y, Z), \\ &= \varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z), \end{aligned} \quad (199)$$

which is three matrix analogue of one half of Corollary 34 and establishes the bound $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z)$ because $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \geq 0$. For the other half of the three matrix analogue of Corollary 34, we can again use Eq. (198) and apply Corollary 34 twice to obtain

$$\begin{aligned} \sum_{\alpha\beta\delta} \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\alpha\beta\delta\phi} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \\ &= \sum_{\alpha\beta\delta\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}^{\beta\delta}(Y, Z))], \\ &= \sum_{\alpha\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \sum_{\beta\delta} \Phi_{\phi}^{\beta\delta}(Y, Z))], \\ &= \sum_{\alpha\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}(Y + Z))], \\ &= \sum_{\alpha\phi} \varphi_{\epsilon}^{\alpha\phi}(X, Y + Z), \\ &= \varphi_{\epsilon}(X + Y + Z), \end{aligned} \quad (200)$$

which in turn establishes the remaining inequality $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\epsilon}(X + Y + Z)$. $\square$

We can now immediately prove the “only if” portion of the claim of Theorem 26. cor:tet_eigenvalue_cubic

Corollary 45. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then for all triples of Young diagrams (α, β, δ) ,*

$$\varphi_{\alpha}(X)^{\frac{2}{3}} \varphi_{\beta}(Y) \varphi_{\delta}(Z)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} \varphi_{\phi}(Y + Z)^{\frac{1}{3}} \varphi_{\gamma}(X + Y)^{\frac{1}{3}} \varphi_{\epsilon}(X + Y + Z)^{\frac{1}{3}}. \quad (201)$$

Proof. The claim follows by summing the inequality in Lemma 43 over triples (γ, ϵ, ϕ) and relaxing both the sides of the resulting inequality. For the lower bound, we have $\varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z) = \sum_{\gamma\epsilon\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ which implies

$$\varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z) \leq \sum_{\gamma\epsilon\phi} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|. \quad (202)$$

Then for the upper bound, Proposition 44 implies

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \leq \min\{\varphi_\alpha(X) \varphi_\phi(Y+Z) \varphi_\gamma(X+Y) \varphi_\delta(Z), \varphi_\epsilon(X+Y+Z)^2, \varphi_\alpha(X)^2 \varphi_\beta(Y)^2 \varphi_\delta(Z)^2\}. \quad (203)$$

Here we elect to remove the minimization by relaxing this inequality to the following:

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \leq [\varphi_\alpha(X) \varphi_\phi(Y+Z) \varphi_\gamma(X+Y) \varphi_\delta(Z) \varphi_\epsilon(X+Y+Z)]^{\frac{2}{3}}. \quad (204)$$

Then the claimed result is obtained by applying Lemma 43 and then dividing by a common factor of $(\varphi_\alpha(X) \varphi_\delta(Z))^{\frac{1}{3}}$ (note that whenever $\varphi_\alpha(X) \varphi_\delta(Z) = 0$, the resulting inequality holds trivially). $\square$

The proof of the “if” portion of Theorem 26 can be found in Section 6.4 which proves the distance bound of Theorem 2 (this implies the “if” portion of Theorem 26 after sending $k \rightarrow \infty$).

5.6 The large n limit & an entropic inequality

sec:large_n

The purpose of this section is to show how to leverage the inverse polynomial bound on $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ for tetrahedral (a, b, c, d, e, f) provided by Corollary 4 to derive inequalities on the entropies of the eigenvalues. Our strategy will be to prove an upper bound on the $6j$ symbol which depends only on the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Corollary 4 (specifically the n -independent bound in Eq. (283)), we will prove the following.

thm:entropic_ineq

Theorem 46. *Let (a, b, c, d, e, f) be tetrahedral eigenvalues. Then*

$$s(c) + s(f) - s(b) - s(e) \geq 0, \quad (205)$$

{eq:entropic_ineq}

where

$$s(x) := - \sum_{i=1}^n x_i \log x_i. \quad (206)$$

{eq:little_s_defn}

We begin with our upper bound on the $6j$ symbol.

lem:sixj_ssa_bound

Lemma 47. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams. Then*

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha, \phi}^\epsilon c_{\beta, \delta}^\phi c_{\alpha, \beta}^\gamma c_{\gamma, \delta}^\epsilon \right)^{\frac{1}{2}} \frac{\dim(V_\gamma^n) \dim(V_\phi^n)}{\dim(V_\beta^n) \dim(V_\epsilon^n)}. \quad (207)$$

Proof. Here we provide a diagrammatic proof of the claimed inequality. For the sake of clarity, we abbreviate the irreducible representation labels, V_λ^n , and omit the multiplicity space labels on the intermediate steps.

$$\begin{aligned} & \left(\epsilon \right)^2 \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix}^\dagger \right] C_{\alpha\phi}^\epsilon C_{\beta\delta}^\phi = \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right] C_{\alpha\beta}^\gamma C_{\gamma\delta}^\epsilon \\ & \leq \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right] C_{\alpha\phi}^\epsilon C_{\beta\delta}^\phi C_{\alpha\beta}^\gamma C_{\gamma\delta}^\epsilon \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right]^\dagger \\ & = \frac{\left(\epsilon \right)^2 \left(\beta \right) \left(\gamma \right) \left(\phi \right)}{\left(\alpha \right) \left(\delta \right)} \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right]^\dagger \left[\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right] \quad (208) \end{aligned}$$

The first equality follows from Definition 39. The inequality follows from an application of the Cauchy-Schwarz inequality with respect to the annotated contraction. The final equality follows from applying Schur’s lemma to components of the diagram mapping V_β^n to itself on the left and V_ϵ^n to itself on the right. Note that the factors of $\dim(V_\epsilon^n)^2$ appearing on left-most and right-most sides of the above inequality cancel producing the claimed inequality. $\square$

An immediate application of Lemma 47 is to combine it with Eq. (283). If (a, b, c, d, e, f) are tetrahedral eigenvalues, then by Corollary 4 there exists a converging sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ such that

$$\frac{1}{q(k)} \leq \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha_k, \phi_k}^{\epsilon_k} c_{\beta_k, \delta_k}^{\phi_k} c_{\alpha_k, \beta_k}^{\gamma_k} c_{\gamma_k, \delta_k}^{\epsilon_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma_k}^n) \dim(V_{\phi_k}^n)}{\dim(V_{\beta_k}^n) \dim(V_{\epsilon_k}^n)}. \quad \text{\texttt{eq:combined_bound}} \quad (209)$$

where $q(k)$ is some polynomial in k independent of the dimension n (see Eq. (283)). Remarkably, the Littlewood-Richardson coefficients $c_{\alpha_k, \phi_k}^{\epsilon_k}$, $c_{\beta_k, \delta_k}^{\phi_k}$, $c_{\alpha_k, \beta_k}^{\gamma_k}$, and $c_{\gamma_k, \delta_k}^{\epsilon_k}$ are also *independent* of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams). This observation provides us with an opportunity to optimize the above bound by varying the value of n .

In particular, it will be useful to better understand how the dimension, $\dim(V_\lambda^n)$, of the irreducible representation V_λ^n of $U(n)$ grows as n becomes much larger than the number of rows of λ . In general, for fixed λ , the function $n \mapsto V_\lambda^n$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_\lambda^n)$ in the regime where n is large compared to $|\lambda|$ and also the last row $\lambda_{\min}$ is large compared to the depth of λ in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, which relates the asymptotics of $\dim(V_\lambda^n)$ to the value of $\dim(W_\lambda)$ (see [Har23, Lemma 2] or [CHW07, Equations (25, 26)]).

prop:harrow_formula

Proposition 48. *Let λ be a Young diagram with depth at most n . Then*

$$\frac{\dim(V_\lambda^n)}{n^{|\lambda|}} = \frac{\dim(W_\lambda)}{|\lambda|!} \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{n} \right). \quad (210)$$

Proof. The statement follows from comparing the Weyl dimension formula for $\dim(V_\lambda^n)$ given in Eq. (211) with the Frobenius dimension formula for $\dim(W_\lambda)$ given in [GW00, Corollaries 9.1.5]:

$$\dim(W_\lambda) = \frac{|\lambda|! \Delta(\lambda + \rho)}{\prod_{i=1}^n (\lambda + \rho)_i!}. \quad \text{\texttt{eq:frob_dim}} \quad (211)$$

That this formula for $\dim(W_\lambda)$ does not depend on n for $n \geq \ell(\lambda)$ and coincides with the Hook-length formula given earlier is proved by [GW00, Corollary 9.1.6]. For additional details, see also [Har23, Appendix A]. $\square$

From Proposition 48 we obtain the following corollary.

cor:harrow_ssa_lim

Corollary 49. *Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that*

$$|\gamma| + |\phi| = |\beta| + |\epsilon|. \quad \text{\texttt{eq:balanced_sizes}} \quad (212)$$

Then

$$\lim_{n \rightarrow \infty} \frac{\dim(V_\gamma^n) \dim(V_\phi^n)}{\dim(V_\beta^n) \dim(V_\epsilon^n)} = \frac{|\beta|! |\epsilon|! \dim(W_\gamma) \dim(W_\phi)}{|\gamma|! |\phi|! \dim(W_\beta) \dim(W_\epsilon)} = \frac{H_\beta H_\epsilon}{H_\gamma H_\phi} \quad (213)$$

Proof. From Proposition 48 one evidently has for any Young diagram λ the limit

$$\lim_{n \rightarrow \infty} \frac{\dim(V_\lambda^n)}{n^{|\lambda|}} = \frac{\dim(W_\lambda)}{|\lambda|!} = \frac{1}{H_\lambda} \quad (214)$$

where H_λ is the product of the hook lengths in λ as defined by Eq. (61). The claim then follows because the missing prefactor, $\frac{n^{|\beta|} n^{|\epsilon|}}{n^{|\gamma|} n^{|\phi|}}$, is equal to one by Eq. (212). $\square$

We are now in a position to prove the main result of this section.

Proof of Theorem 46. As (a, b, c, d, e, f) are tetrahedral, the inverse polynomial bound on $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ for tetrahedral (a, b, c, d, e, f) provided by Corollary 4 implies it must at least be non-zero and thus Eq. (212)

holds. Therefore, by Corollary 49 and the discussion surrounding the bound in Eq. (209) we conclude that along the convergent sequence of Young diagrams,

$$\lim_{k \rightarrow \infty} \frac{1}{k} \log \left(\frac{H_{\beta_k} H_{\epsilon_k}}{H_{\gamma_k} H_{\phi_k}} \right) \geq 0 \quad \text{\textcolor{blue}{\{eq:hook_ratio_limit\}}} \quad (215)$$

provided the limit. To prove the limit exists, an application of Stirling's approximation, $\log n! \approx n \log n - n$ [Rob55], to the approximation formula $H_\lambda \approx \prod_{i=1}^n \lambda_i!$ produces for any constant η the formula

$$\frac{1}{\eta} \log H_\lambda \approx (\log \eta - 1)|\lambda| + \sum_{i=1}^n \frac{\lambda_i}{\eta} \log \frac{\lambda_i}{\eta} \quad (216)$$

Because $|\gamma| + |\phi| = |\beta| + |\epsilon|$ (Eq. (212)), the $(\log \eta - 1)$ terms will cancel in Eq. (215). Then using the fact that that $(\beta_k, \gamma_k, \epsilon_k, \phi_k)$ converge in proportion to the eigenvalues (b, c, e, f) (with proportionality constant $\eta = \frac{k}{\text{Tr}[e]}$), we conclude that Eq. (215) has a limit and moreover it equals the left-hand side of Eq. (205). $\square$

TC: THE FOLLOWING SECTION IS CURRENTLY BEING RE-REWRITTEN INTO PREVIOUS SECTIONS

6 Distance bounds via eigenvalue concentration

[sec:distance_bounds](#)

The purpose of this section is to provide a proof of the distance bound in Theorem 2. With Theorem 2 established, the “if” portion of Theorem 26 follows immediately. Note that “only if” portion of Theorem 26 is just a restatement of Corollary 45 proven in the previous sections.

These bounds make use of eigenvalue concentration techniques, which we review first in Section 6.1, as well as formulas from random matrix theory, which we review afterwards in Section 6.2. As a warm-up exercise, in Section 6.3 we demonstrate how these techniques can be applied to obtain a distance bound for the original Horn problem. Then we generalize the proof technique in Section 6.4 to prove Theorem 2.

6.1 Eigenvalue estimation & separation

[sec:eig_est_and_sep](#)

In this section we show how the representations $\pi_\lambda(X)$ of Hermitian matrices X encode useful information about the eigenvalues of X . These results apply to the case where $X \geq 0$ is a positive semidefinite matrix, in which case the representation $\pi_\lambda(X) \geq 0$ is positive semidefinite also. Thus whenever $X \geq 0$, one can define a probability distribution known as the *Schur-Weyl* distribution [OW16].

Definition 50. The Schur-Weyl distribution of order k for a positive semidefinite matrix $X \geq 0$ with eigenvalues $x = (x_1, \dots, x_n)$ is defined as the probability distribution over Young diagrams λ with k boxes with probability

$$p_x(\lambda) := \frac{\text{Tr}[P_\lambda X^{\otimes k}]}{\text{Tr}[X^{\otimes k}]} = \frac{\dim W_\lambda s_\lambda(x)}{\text{Tr}[x]^k} = \frac{|\lambda|!}{\text{Tr}[x]^k} \varphi_\lambda(x). \quad (\text{eq:schur_weyl}) \quad (217)$$

A key property of the Schur-Weyl distribution $p_x(\lambda)$ is that it behaves similarly to the multinomial distribution $m_x(\lambda)$ where

$$m_x(\lambda) := \frac{|\lambda|!}{\text{Tr}[x]^k} \frac{x_1^{\lambda_1}}{\lambda_1!} \frac{x_2^{\lambda_2}}{\lambda_2!} \cdots \frac{x_n^{\lambda_n}}{\lambda_n!}. \quad (218)$$

[prop:multinomial_bound](#)

Proposition 51. Given a Young diagram λ and non-negative eigenvalues x ,

$$p_x(\lambda) \leq \dim V_\lambda^n m_x(\lambda). \quad (219)$$

Proof. The proof follows from (i) an upper bound on the Schur-polynomial $s_\lambda(x)$,

$$s_\lambda(x) \leq \dim V_\lambda^n x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}, \quad (220)$$

which follows because $s_\lambda(x)$ can be expressed as a sum over $\dim V_\lambda^n$ multinomials of the form $x_1^{\mu_1} x_2^{\mu_2} \cdots x_n^{\mu_n}$ with $x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}$ the largest, and (ii) a lower bound on the Hook-product,

$$H_\lambda \geq \lambda_1! \cdots \lambda_n!, \quad (221)$$

which follows because $h_\lambda(b) \geq \lambda_i - j + 1$ holds for the box b in row i and column j . $\square$

Remark 52. We will henceforth relax the dimension prefactor $\dim V_\lambda^n$ using the following dimension bound which holds for all Young diagrams with k boxes:

$$\dim V_\lambda^n \leq (k+1)^{\binom{n}{2}} := (k+1)^{\frac{n(n-1)}{2}}. \quad (\text{eq:unitary_irrep_dim_bound}) \quad (222)$$

The following is known result is known as eigenvalue or spectrum estimation [ARS88; KW01; HM02; CM06; OW16].

Proposition 53 (Eigenvalue estimation). *Let $X \geq 0$ be a positive semidefinite $n \times n$ matrix with eigenvalues $x = (x_1 \geq \dots \geq x_n \geq 0)$. Then the Schur-Weyl distribution $p_x(\lambda)$ for $\lambda \vdash k$ satisfies*

$$p_x(\lambda) \leq (k+1)^{\binom{n}{2}} \exp \left(-kD \left(\frac{\lambda}{k} \parallel \frac{x}{\text{Tr}[x]} \right) \right) \quad (223)$$

where $D(p||q)$ denotes the Kullback-Leibler divergence of a distribution q from a distribution p :

$$D(p||q) = \sum_{i=1}^n p_i \log \frac{p_i}{q_i}. \quad (224)$$

The Kullback-Leibler divergence, $D(p||q)$, is a non-negative quantity which vanishes if and only if $p = q$. Consequently, eigenvalue estimation captures the fact that, like the multinomial distribution $m_x(\lambda)$, the Schur-Weyl distribution $p_x(\lambda)$ is sharply concentrated around Young diagrams λ whose distribution, λ/k , is similar to the eigenvalue distribution $x/\text{Tr}[x]$.

For our purposes, it will instead be useful to derive a result which compares the eigenvalues of a pair of positive semidefinite matrices $X, Y \geq 0$ directly. Recall that the *Bhattacharyya coefficient* $\text{BC}(p, q) \in [0, 1]$ between two probability distributions p and q over a common set $\{1, \dots, r\}$ is defined by $\text{BC}(p, q) = \sum_{i=1}^r \sqrt{p_i q_i}$ and is equal one if and only if $p = q$. For notational convenient, we will extend this definition of Bhattachara coefficient to consider unnormalized distributions in the following way. Suppose $p = (p_1, \dots, p_r)$ and $q = (q_1, \dots, q_r)$ are non-negative tuples over the set $\{1, \dots, r\}$ which are not necessarily normalized, but nevertheless have the same total $\tau = \sum_{i=1}^r p_i = \sum_{i=1}^r q_i$. Then we define the *unnormalized Bhattachara coefficient* $\text{BC}(p, q) \in [0, \tau]$ between p and q by

$$\text{BC}(p, q) = \sum_{i=1}^r \sqrt{p_i q_i}, \quad (225)$$

where $\text{BC}(p, q) = \tau$ if and only if $p = q$. The unnormalized Bhattachara coefficient inherits all of its properties from the normalized Bhattachara coefficient by homogeneity:

$$\text{BC}(p, q) = \tau \text{BC} \left(\frac{p}{\tau}, \frac{q}{\tau} \right). \quad (226)$$

Lemma 54 (Eigenvalue separation). *Let x and y non-negative eigenvalues of length n with the same trace $\tau = \text{Tr}[x] = \text{Tr}[y]$. For any non-negative integer $k \geq 1$,*

$$\sum_{\lambda \vdash k} \sqrt{\varphi_\lambda(x) \varphi_\lambda(y)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(x, y)^k}{k!}, \quad (227)$$

where $\text{BC}(x, y)$ is unnormalized Bhattacharyya coefficient between the eigenvalues x and y and $\text{BC}(x, y) = \tau$ if and only if $x = y$.

Proof. We may assume that x, y are non-zero and thus $\tau > 0$. The left-hand side of Eq. (227) is the equal to the unnormalized Bhattacharyya coefficient between $\varphi_\lambda(x)$ and $\varphi_\lambda(y)$. As $\varphi_\lambda(x)$ is proportional to the Schur-Weyl distribution $p_x(\lambda)$ by definition (Eq. (217)), we have

$$\sum_{\lambda \vdash k} \sqrt{\varphi_\lambda(x) \varphi_\lambda(y)} = \frac{\tau^k}{k!} \sum_{\lambda \vdash k} \sqrt{p_x(\lambda) p_y(\lambda)} = \frac{\tau^k}{k!} \text{BC}_k(p_x, p_y) \quad (228)$$

where $\text{BC}_k(p_x, p_y)$ denotes the normalized Bhattacharyya coefficient between the Schur-Weyl distributions p_x and p_y over Young diagrams of size k . From Proposition 51 and Eq. (222), we have

$$\text{BC}_k(p_x, p_y) \leq (k+1)^{\binom{n}{2}} \text{BC}_k(m_x, m_y). \quad (229)$$

where $\text{BC}_k(m_x, m_y)$ is the Bhattacharyya coefficient between the multinomial distributions m_x and m_y . Since Bhattacharyya coefficients are multiplicative with respect to multinomial distributions, i.e.,

$$\text{BC}_k(m_x, m_y) = \text{BC}\left(\frac{x}{\tau}, \frac{y}{\tau}\right)^k = \frac{\text{BC}(x, y)^k}{\tau^k}, \quad (230)$$

we conclude

$$\sum_{\lambda \vdash k} \sqrt{\varphi_\lambda(x) \varphi_\lambda(y)} \leq \frac{\tau^k}{k!} (k+1) \binom{n}{2} \frac{\text{BC}(x, y)^k}{\tau^k} \quad (231)$$

which is equal to Eq. (227). $\square$

In subsequent sections we will frequently use of the *weaker* bound

$$\forall \lambda \vdash k : \quad \sqrt{\varphi_\lambda(x) \varphi_\lambda(y)} \leq (k+1) \binom{n}{2} \frac{\text{BC}(x, y)^k}{k!}, \quad \text{\texttt{\{eq:weak_eig_sep\}}} \quad (232)$$

which follows from Lemma 54 because $\varphi_\lambda(x), \varphi_\lambda(y) \geq 0$ whenever the eigenvalues x, y are non-negative.

Remark 55 (Distances between distributions). In order to quantify the separation between distributions more geometrically, consider the Kolmogorov (or total variation) distance between two normalized probability distributions p and q :

$$K(p, q) := \frac{1}{2} \|p - q\|_1. \quad (233)$$

The normalized Bhattacharyya coefficient $\text{BC}(p, q)$ and Kolmogorov distance $K(p, q)$ are related by the inequalities

$$1 - \text{BC}(p, q) \leq K(p, q) \leq \sqrt{1 - \text{BC}(p, q)^2} \quad (234)$$

For a proof, see for example [FV02, Proposition 5, Equation (41)]. For the purposes of this paper, we only need the bound $K(p, q) \leq \sqrt{1 - \text{BC}(p, q)^2}$ which is equivalent to $\text{BC}(p, q) \leq \sqrt{1 - K(p, q)^2}$ which can be relaxed to obtain⁶

$$\text{BC}(p, q) \leq \exp\left(-\frac{1}{2} K(p, q)^2\right) = \exp\left(-\frac{1}{8} \|p - q\|_1^2\right). \quad (235)$$

Stated in terms of the unnormalized Bhattacharyya coefficient $\text{BC}(p, q)$ between unnormalized distributions p, q with the same total τ , this inequality is equivalent to

$$\frac{\text{BC}(p, q)}{\tau} \leq \exp\left(-\frac{1}{8\tau^2} \|p - q\|_1^2\right). \quad \text{\texttt{\{eq:Bhattacharyya_vs_distance\}}} \quad (236)$$

In subsequent sections we will use Eq. (236) to convert lower bounds on Bhattacharyya coefficients $\text{BC}(p, q)$ into upper bounds on distances $\|p - q\|_1$.

6.2 Results for random matrices

[sec:random_matrices](#)

The purpose of this section is record some results concerning sums of random Hermitian matrices with fixed eigenvalues sampled uniformly from their respective orbits. The first two propositions follow from standard applications of Schur's lemma, and are used to prove the others.

Proposition 56. *Let $x = (x_1 \geq \dots \geq x_n)$ be a set of eigenvalues and let $X \sim \mathcal{O}_x$ denote a random Hermitian matrix of the form $X = U[x]U^{-1}$ where $U \in \text{U}(n)$ is sampled uniformly with respect to the normalized Haar measure on $\text{U}(n)$. Then for any Young diagram λ ,*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \pi_\lambda(X) = s_\lambda(x) \frac{\text{id}_{V_\lambda^n}}{\dim V_\lambda^n}. \quad (237)$$

⁶Here we used the inequality $\sqrt{1 - x^2} \leq \exp(-x^2/2)$ which holds for all $x \in [0, 1]$.

Corollary 57. *Let (x, y) be a pair of eigenvalues. Then*

$$\mathbb{E}_{\substack{(X,Y) \sim \\ \mathcal{O}_x \times \mathcal{O}_y}} \frac{\varphi_\gamma^{\alpha\beta}(X, Y)}{\dim V_\gamma^n} = c_{\alpha\beta}^\gamma \frac{\varphi_\alpha(x)}{\dim V_\alpha^n} \frac{\varphi_\beta(y)}{\dim V_\beta^n}. \quad (238)$$

cor:random_matrices_tet

Corollary 58. *Let (x, y, z) be a triple of eigenvalues. Then*

$$\mathbb{E}_{\substack{(X,Y,Z) \sim \\ \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z}} \frac{\varphi_{\gamma \in \phi}^{\alpha\beta\delta}(X, Y, Z)}{\dim V_\epsilon^n} = \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \frac{\varphi_\alpha(x)}{\dim V_\alpha^n} \frac{\varphi_\beta(y)}{\dim V_\beta^n} \frac{\varphi_\delta(z)}{\dim V_\delta^n}. \quad (239)$$

6.3 Distance bound for the original Horn problem

sec:dist_bound_for_orig_horn

In this section, we prove the following proposition pertaining to the original Horn problem as a warm-up exercise before we prove its generalization for the tetrahedral Horn problem (Theorem 2) in Section 6.4.

prop:approx_tri

Proposition 59. *Let (a, b, c) be a triple of non-negative eigenvalues satisfying trace condition $\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c]$. If for all pairs of Young diagrams (α, β) of total size $|\alpha| + |\beta| = k$,*

$$\varphi_\alpha(a) \varphi_\beta(b) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\gamma(c), \quad (240)$$

feq:tri_ab_c_nec_ineq

then

$$\frac{D(a, b; c)}{\text{Tr}[c]} \leq O \left(n \sqrt{\frac{\log k}{k}} \right). \quad (241)$$

feq:tri_ab_c_dist_bound

where $D(a, b; c) := \min_{(A,B) \in \mathcal{O}_a \times \mathcal{O}_b} \|\text{eig}(A + B) - c\|_1$.

Proof of Proposition 59. The proof proceeds in two stages. In the first stage we fix an arbitrary pair (α, β) with total size $|\alpha| + |\beta| = k$ and derive multiple bounds on the product $\varphi_\alpha(a) \varphi_\beta(b)$. Then in the second stage, we sum over all such pairs and then obtain the claimed bound on $D(a, b; c)$.

To begin, we will bound $\varphi_\alpha(a) \varphi_\beta(b)$ in two ways and then apply the eigenvalue separation lemma (specifically Eq. (232)). The first bound on $\varphi_\alpha(a) \varphi_\beta(b)$ is obtained from relaxing the assumption in Eq. (240) by considering a Young diagram γ^* of size k which maximizes $\varphi_{\gamma^*}(c)$ among those γ such that $c_{\alpha\beta}^\gamma > 0$ for the given (a, b, c) . As there are at most $(k+1)^{n-1}$ Young diagrams of size k , we conclude

$$\varphi_\alpha(a) \varphi_\beta(b) \leq (k+1)^{n-1} \varphi_{\gamma^*}(c). \quad (242)$$

feq:first_ab_bound

The second bound on $\varphi_\alpha(a) \varphi_\beta(b)$ is obtained by considering Corollary 57 which states that for any γ such that $c_{\alpha\beta}^\gamma > 0$,

$$\varphi_\alpha(a) \varphi_\beta(b) = \frac{\dim V_\alpha^n \otimes V_\beta^n}{c_{\alpha\beta}^\gamma \dim V_\gamma^n} \mathbb{E}_{(X,Y) \sim \mathcal{O}_x \times \mathcal{O}_y} \varphi_\gamma^{\alpha\beta}(X, Y). \quad (243)$$

To relax this into a usable bound we (i) bound the dimensions by $\frac{\dim V_\alpha^n \otimes V_\beta^n}{c_{\alpha\beta}^\gamma \dim V_\gamma^n} \leq \dim(V_\alpha^n \otimes V_\beta^n) \leq (k+1)^{2\binom{n}{2}}$, then (ii) bound the average over $X \in \mathcal{O}_x$ and $Y \in \mathcal{O}_y$ with a maximum, then (iii) apply Corollary 37 to get $\varphi_\gamma^{\alpha\beta}(X, Y) \leq \varphi_\gamma(X + Y)$ and then finally (iv) specialize to the optimal $\gamma \mapsto \gamma^*$ to produce

$$\varphi_\alpha(a) \varphi_\beta(b) \leq (k+1)^{2\binom{n}{2}} \max_{\substack{(X,Y) \in \\ \mathcal{O}_x \times \mathcal{O}_y}} \varphi_{\gamma^*}(X + Y). \quad (244)$$

feq:second_ab_bound

Then we combine the two bounds on $\varphi_\alpha(a)\varphi_\beta(b)$ from Eq. (242) and Eq. (244) and apply the eigenvalue separation bound Eq. (232) to obtain

$$\varphi_\alpha(a)\varphi_\beta(b) \leq \sqrt{(k+1)^{2\binom{n}{2}}(k+1)^{n-1} \max_{\substack{(X,Y) \in \\ \mathcal{O}_a \times \mathcal{O}_b}} \varphi_{\gamma^*}(X+Y)\varphi_{\gamma^*}(c)}, \quad (245)$$

$$\leq \sqrt{(k+1)^{2\binom{n}{2}}(k+1)^{n-1}(k+1)^{\binom{n}{2}} \max_{\substack{(X,Y) \in \\ \mathcal{O}_a \times \mathcal{O}_b}} \frac{\text{BC}(\text{eig}(X+Y), c)^k}{k!}}, \quad (246)$$

$$= (k+1)^{2\binom{n}{2} + \frac{n-1}{2}} \frac{\text{BC}(a, b; c)^k}{k!}, \quad \text{\texttt{\{eq:ab_intermediate_bound\}}} \quad (247)$$

where $\text{BC}(a, b; c)$ is defined as the maximum Bhattacharyya coefficient

$$\text{BC}(a, b; c) := \max_{\substack{(X,Y) \in \\ \mathcal{O}_a \times \mathcal{O}_b}} \text{BC}(\text{eig}(X+Y), c). \quad (248)$$

Now we sum both sides of Eq. (247) over *all* pairs (α, β) with total size $|\alpha| + |\beta| = k$ to obtain

$$\frac{\text{Tr}[c]^k}{k!} \leq (k+1)^{2\binom{n}{2} + \frac{n-1}{2} + 2n} \frac{\text{BC}(a, b; c)^k}{k!}, \quad (249)$$

where the left-hand side follows from the trace condition $\text{Tr}[c] = \text{Tr}[a] + \text{Tr}[b]$ and

$$\sum_{\ell=0}^k \sum_{\substack{\alpha \vdash \ell \\ \beta \vdash k-\ell}} \varphi_\alpha(a)\varphi_\beta(b) = \sum_{\ell=0}^k \frac{\text{Tr}[a]^\ell}{\ell!} \frac{\text{Tr}[b]^{k-\ell}}{(k-\ell)!} = \frac{(\text{Tr}[a] + \text{Tr}[b])^k}{k!}, \quad \text{\texttt{\{eq:binomial_trace\}}} \quad (250)$$

whereas the extra factor of $2n$ in the exponent on the right-hand side is obtained because there are at most $(k+1)^n$ Young diagrams with n rows and size at most k and we are summing over a pair (α, β) with total size $|\alpha| + |\beta| = k$. This establishes a lower bound on $\text{BC}(a, b; c)$ of the form

$$\frac{\text{BC}(a, b; c)}{\text{Tr}[c]} \geq \left(\frac{1}{(k+1)^{\frac{2n^2+3n-1}{2}}} \right)^{\frac{1}{k}} \quad (251)$$

Finally, using Eq. (236), we substitute the maximum normalized Bhattacharyya coefficient $\frac{\text{BC}(a, b; c)}{\text{Tr}[c]}$ with a minimum normalized distance to obtain Eq. (241) which completes the proof. $\square$

6.4 Distance bound for the tetrahedral Horn problem

[sec:dist_bound_for_tet_horn](#)

The purpose of this section is to prove our main distance bound, Theorem 2. Note that Theorem 2 then immediately implies the “if” portion of the claim of Theorem 26 by taking $k \rightarrow \infty$. The “only if” portion of Theorem 26 was proven in Corollary 45.

Proof of Theorem 2. The proof of Theorem 2 mimics the structure of the proof of Proposition 59. In the first stage, we fix an arbitrary triple (α, β, δ) with total size $|\alpha| + |\beta| + |\delta| = k$ and derive multiple bounds on the product $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$. Then in the second stage, we sum over all such triples (α, β, δ) and obtain the bound on $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$.

We begin by bounding $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ in two separate ways and then afterward combine them to apply the eigenvalue separation lemma (specifically Eq. (232)). The first bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ is obtained from relaxing the assumption of the claim that (a, b, c, d, e, f) satisfies Eq. (105) for all triples of Young diagrams (α, β, δ) . By multiplying both sides of Eq. (105) by $\varphi_\alpha(a)^{\frac{1}{3}}\varphi_\delta(d)^{\frac{1}{3}}$ we conclude

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\delta(d)\varphi_\epsilon(e))^{\frac{1}{3}}. \quad \text{\texttt{\{eq:abd_multiplied\}}} \quad (252)$$

Now let $(\gamma^*, \epsilon^*, \phi^*)$ denote a triple of Young diagrams which maximizes the summand above. Note that the summand is only non-zero if the size constraints $|\alpha| + |\beta| + |\delta| = |\alpha| + |\phi| = |\gamma| + |\delta| = |\epsilon| = k$ are satisfied. As there are at most $(k+1)^{3(n-1)}$ such triples satisfying the size constraints (remember α, β, δ are fixed), we conclude that

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{3(n-1)} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{smallmatrix} \right\} \right\|_\infty (\varphi_\alpha(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_\delta(d)\varphi_{\epsilon^*}(e))^{\frac{1}{3}}. \quad (\text{eq:first_abd_bound}) \quad (253)$$

The second bound is obtained by considering Corollary 58 which states that for any triple (γ, ϵ, ϕ) we have

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) = \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \mathbb{E}_{(X,Y,Z) \sim \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \quad (254)$$

To obtain our desired bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ we make a series of relaxations. First we bound the dimensions of the representations by $\frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \leq \dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n) \leq (k+1)^{3\binom{n}{2}}$. Then we upper bound the average of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over $(X, Y, Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d$ with the maximum of the absolute value. Together this gives

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{3\binom{n}{2}} \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|, \quad (\text{eq:max_abs_value_bound}) \quad (255)$$

Furthermore, applying Lemma 43 and Proposition 44 to the absolute value $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$ on the right-hand side above produces

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty (\varphi_\alpha(X)\varphi_\phi(Y+Z)\varphi_\gamma(X+Y)\varphi_\delta(Z)\varphi_\epsilon(X+Y+Z))^{\frac{1}{3}}. \quad (\text{eq:sup_bound}) \quad (256)$$

Applying Eq. (256) to Eq. (255) gives

$$\begin{aligned} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &\leq (k+1)^{3\binom{n}{2}} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty \times \dots \\ &\dots \times \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} (\varphi_\alpha(X)\varphi_\phi(Y+Z)\varphi_\gamma(X+Y)\varphi_\delta(Z)\varphi_\epsilon(X+Y+Z))^{\frac{1}{3}}, \end{aligned} \quad (\text{eq:second_abd_bound}) \quad (257)$$

Now we choose (γ, ϵ, f) in the above bound to be equal to the optimal choice from $(\gamma^*, \epsilon^*, \phi^*)$ from Eq. (253) and multiply the result with Eq. (253) and then take a square root to obtain

$$\begin{aligned} \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &\leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1)} \times \dots \\ &\dots \times \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} (\varphi_\alpha(a)\varphi_\alpha(X)\varphi_{\phi^*}(f)\varphi_{\phi^*}(Y+Z)\varphi_{\gamma^*}(c)\varphi_{\gamma^*}(X+Y)\varphi_\delta(d)\varphi_\delta(Z)\varphi_{\epsilon^*}(e)\varphi_{\epsilon^*}(X+Y+Z))^{\frac{1}{6}}. \end{aligned} \quad (\text{eq:third_abd_bound}) \quad (258)$$

Note we have eliminated the $6j$ symbols by using the bound $\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty^2 \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2$. We are now in a position to apply the eigenvalue separation lemma (specifically Eq. (232)). For each Young diagram $\lambda \in \{\alpha, \delta, \gamma, \epsilon, \phi\}$ we apply Eq. (232) (noting $|\lambda| \leq k$ for each) to conclude

$$\sqrt{\varphi_\alpha(a)\varphi_\alpha(X)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(a, \text{eig}(X))^{| \alpha |}}{| \alpha |!} \quad (259)$$

$$\sqrt{\varphi_\delta(d)\varphi_\delta(Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(d, \text{eig}(Z))^{| \delta |}}{| \delta |!} \quad (260)$$

$$\sqrt{\varphi_\gamma(c)\varphi_\gamma(X+Y)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(c, \text{eig}(X+Y))^{| \gamma |}}{| \gamma |!}, \quad (261)$$

$$\sqrt{\varphi_\phi(f)\varphi_\phi(Y+Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(f, \text{eig}(Y+Z))^{| \phi |}}{| \phi |!}, \quad (262)$$

$$\sqrt{\varphi_\epsilon(e)\varphi_\epsilon(X+Y+Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(e, \text{eig}(X+Y+Z))^{| \epsilon |}}{| \epsilon |!}. \quad (263)$$

At this stage, we can combine the bounds for α and ϕ to obtain an expression involving the same power of k :

$$\sqrt{\varphi_\alpha(a)\varphi_\alpha(X)}\sqrt{\varphi_\phi(f)\varphi_\phi(Y+Z)} \leq (k+1)^{n(n-1)} \frac{\text{BC}(a, \text{eig}(X))^{| \alpha |}}{| \alpha |!} \frac{\text{BC}(f, \text{eig}(Y+Z))^{| \phi |}}{| \phi |!}, \quad (264)$$

$$\leq (k+1)^{n(n-1)} \frac{[\text{BC}(a, \text{eig}(X)) + \text{BC}(f, \text{eig}(Y+Z))]^k}{k!}, \quad (265)$$

$$= (k+1)^{n(n-1)} \frac{\text{BC}(a \oplus f, \text{eig}(X \oplus Y + Z))^k}{k!}, \quad (266)$$

because $| \alpha | + | \phi | = | \epsilon | = k$. A similar bound also follows from combining γ with δ . Define the maximum product Bhattacharyya coefficient as

$$\text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \max_{\substack{(X,Y,Z) \in \\ \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} \text{BC}(a \oplus f, \text{eig}(X \oplus Y + Z)) \text{BC}(c \oplus d, \text{eig}(X + Y \oplus Z)) \text{BC}(e, \text{eig}(X + Y + Z)). \quad (267)$$

From Eq. (258) and the above Bhattacharyya coefficient bounds we obtain

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1) + \frac{3}{3}n(n-1)} \frac{1}{k!} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{k}{3}} \quad (268)$$

Now that we have established an upper bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ which is uniform in the triple (α, β, δ) , the next step is to sum this expression over all triples (α, β, δ) such that $| \alpha | + | \beta | + | \delta | = k$. Note that summing $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ over (α, β, δ) with total size $| \alpha | + | \beta | + | \delta | = k$ yields the analogue of Eq. (250):

$$\sum_{\substack{\alpha, \beta, \delta \\ | \alpha | + | \beta | + | \delta | = k}} \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) = \frac{(\text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d])^k}{k!} = \frac{\text{Tr}[e]^k}{k!} \quad (269)$$

where we used that the trace conditions hold, i.e., $\text{Tr}[e] = \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d]$.

As there are at most $(k+1)^{3n}$ triples Young diagrams (α, β, δ) with total size $| \alpha | + | \beta | + | \delta | = k$ being summed here, we finally conclude

$$\frac{\text{Tr}[e]^k}{k!} \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1) + \frac{3}{3}n(n-1) + 3n} \frac{1}{k!} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{k}{3}} \quad (270)$$

or equivalently

$$\frac{1}{\text{Tr}[e]} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{1}{3}} \geq \left(\frac{1}{(k+1)^{\frac{7n^2+11n+6}{4}}} \right)^{\frac{1}{k}}. \quad (271)$$

Finally, using Eq. (236), we can substitute the maximum normalized product of Bhattacharyya coefficients here with minimum normalized sum of distances (as in Eq. (101)) to obtain Eq. (106) which completes the proof. $\square$

6.5 Asymptotics of the $6j$ symbols

In order to prove Corollary 4 it will be useful to have a bound on the quantity $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$ which directly relates the sextuple of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to the eigenvalues of $(X, Y, X+Y, Z, X+Y+Z, Y+Z)$. *lem:asym_sizej_helper*

Lemma 60. *To begin let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices and let (a, b, c, d, e, f) be the eigenvalues of the matrices $(X, Y, X+Y, Z, X+Y+Z, Y+Z)$. Then for all sextuples of Young diagrams, $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$, with sizes satisfying*

$$k = | \epsilon | = | \alpha | + | \beta | + | \delta | = | \alpha | + | \phi | = | \gamma | + | \delta |, \quad (272) \quad \text{\small *\{eq:yd_k_sizes\}}*$$

we have (assuming $\text{Tr}[e] \neq 0$) the following bounds

$$\frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty}, \quad \text{\texttt{eq:hex_6j_bound}} \quad (273)$$

$$\frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq (k+1)^{n(n-1)} \exp \left(-\frac{k}{4} R_{abcdef}^{\alpha\beta\gamma\delta\epsilon\phi} \right), \quad \text{\texttt{eq:hex_rel_ent_bound}} \quad (274)$$

where

$$R_{abcdef}^{\alpha\beta\gamma\delta\epsilon\phi} = D \left(\frac{\alpha \oplus \beta \oplus \delta}{k} \left\| \frac{a \oplus b \oplus d}{\text{Tr}[e]} \right\| \right) + D \left(\frac{\gamma \oplus \delta}{k} \left\| \frac{c \oplus d}{\text{Tr}[e]} \right\| \right) + D \left(\frac{\alpha \oplus \phi}{k} \left\| \frac{a \oplus f}{\text{Tr}[e]} \right\| \right) + D \left(\frac{\epsilon}{k} \left\| \frac{e}{\text{Tr}[e]} \right\| \right). \quad (275)$$

Proof. A combination of Lemma 43 and Proposition 44 implies

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} (\varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d) \times \varphi_{\gamma}(c)\varphi_{\delta}(d) \times \varphi_{\epsilon}(e) \times \varphi_{\alpha}(a)\varphi_{\phi}(f))^{\frac{1}{4}}, \quad \text{\texttt{eq:abs_quad_bound}} \quad (276)$$

From here we note that each of $\varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d)$, $\varphi_{\gamma}(c)\varphi_{\delta}(d)$, $\varphi_{\epsilon}(e)$, and $\varphi_{\alpha}(a)\varphi_{\phi}(f)$ is upper bounded by $\frac{\text{Tr}[e]^k}{k!}$ because they each sum to $\frac{\text{Tr}[e]^k}{k!}$ when summing over the corresponding sets of Young diagrams with sizes given by Eq. (272) (see Eq. (250)). Applying these upper bounds to Eq. (276) (and multiplying by $\frac{k!}{\text{Tr}[e]^k}$) gives Eq. (273).

For the proof if Eq. (274), consider the relative entropy bounds arising from eigenvalue estimation Proposition 53:

$$\frac{k!}{\text{Tr}[e]^k} \varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d) \leq (k+1)^{3\binom{n}{2}} \exp \left[-kD \left(\frac{\alpha \oplus \beta \oplus \delta}{k} \left\| \frac{a \oplus b \oplus d}{\text{Tr}[e]} \right\| \right) \right], \quad (277)$$

$$\frac{k!}{\text{Tr}[e]^k} \varphi_{\gamma}(c)\varphi_{\delta}(d) \leq (k+1)^{2\binom{n}{2}} \exp \left[-kD \left(\frac{\gamma \oplus \delta}{k} \left\| \frac{c \oplus d}{\text{Tr}[e]} \right\| \right) \right], \quad (278)$$

$$\frac{k!}{\text{Tr}[e]^k} \varphi_{\alpha}(a)\varphi_{\phi}(f) \leq (k+1)^{2\binom{n}{2}} \exp \left[-kD \left(\frac{\alpha \oplus \phi}{k} \left\| \frac{a \oplus f}{\text{Tr}[e]} \right\| \right) \right], \quad (279)$$

$$\frac{k!}{\text{Tr}[e]^k} \varphi_{\epsilon}(e) \leq (k+1)^{\binom{n}{2}} \exp \left[-kD \left(\frac{\epsilon}{k} \left\| \frac{e}{\text{Tr}[e]} \right\| \right) \right]. \quad (280)$$

Applying these bounds to Eq. (276) and using $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} \leq 1$, we obtain Eq. (274) $\square$

Proof of Corollary 4. First we prove the claimed inverse polynomial lower bound, and then afterwards prove the exponential decay.

Assume (a, b, c, d, e, f) are the eigenvalues of positive semidefinite matrices (A, B, C, D, E, F) satisfying the tetrahedral constraints. Summing $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$ over all sextuples of Young diagrams satisfying Eq. (272) produces

$$\sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon|=k}} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) = \frac{\text{Tr}[e]^k}{k!}. \quad (281)$$

Therefore, the sextuple which maximizes the absolute value of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$ over this k -dependent domain of summation, which we denote by $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, must satisfy

$$\frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma_k\epsilon_k\phi_k}^{\alpha_k\beta_k\delta_k}(A, B, D) \right| \geq \frac{1}{(k+1)^{6\text{rank}[e]}} \quad (282)$$

where $(k+1)^{6\text{rank}[e]}$ is a crude upper bound on the number of terms in the above summation which are non-zero as there are at most $(k+1)^r$ Young diagrams with size at most k and depth at most r (note that each of $\varphi_{\alpha}(a), \varphi_{\beta}(b), \varphi_{\delta}(d), \varphi_{\phi}(f), \varphi_{\gamma}(c)$ and $\varphi_{\epsilon}(e)$ are respectively only non-zero if the depth of the

corresponding Young diagram is at most the rank of the corresponding set of eigenvalues and $\text{rank}[e]$ is always the maximum possible rank). Therefore, applying Eq. (273) in Lemma 60 gives

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty \geq \frac{1}{(k+1)^{6 \text{rank}[e]}}, \quad \text{\textcolor{blue}{\{eq:n_independent_inverse_poly\}}} \quad (283)$$

while Eq. (274) in Lemma 60 gives

$$\exp\left(-\frac{k}{4} R_{abcdef}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k}\right) \geq \frac{1}{(k+1)^{n(n-1)}} \frac{1}{(k+1)^{6 \text{rank}[e]}}. \quad (284)$$

Thus we have demonstrated the existence of a sequence of sextuples of Young diagrams for which (i) $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty$ satisfies the claimed inverse polynomial lower bound, and also (ii) $R_{abcdef}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \rightarrow 0$ which implies the convergence claim:

$$\frac{\alpha_k}{k} \rightarrow \frac{a}{\text{Tr}[e]}, \quad \frac{\beta_k}{k} \rightarrow \frac{b}{\text{Tr}[e]}, \quad \frac{\gamma_k}{k} \rightarrow \frac{c}{\text{Tr}[e]}, \quad \frac{\delta_k}{k} \rightarrow \frac{d}{\text{Tr}[e]}, \quad \frac{\epsilon_k}{k} \rightarrow \frac{e}{\text{Tr}[e]}, \quad \frac{\phi_k}{k} \rightarrow \frac{f}{\text{Tr}[e]}. \quad (285)$$

We now prove the exponential decay on $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty$ when $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ converges to some non-tetrahedral eigenvalues. It will be slightly easier to prove the exponential decay of $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2$ which is actually a bit stronger than necessary. Without loss of generality, we can assume the Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ satisfy the size constraints Eq. (272) as before (since otherwise $\begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} = 0$). Then from Corollary 58, we conclude that for *any* eigenvalues (a, b, d) ,

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \varphi_{\alpha_k}(a) \varphi_{\beta_k}(b) \varphi_{\delta_k}(d) = \frac{\dim(V_{\alpha_k}^n \otimes V_{\beta_k}^n \otimes V_{\delta_k}^n)}{\dim(V_{\epsilon_k}^n)} \mathbb{E}_{\substack{(X,Y,Z) \sim \\ \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z). \quad (286)$$

As in the proof of Theorem 2 we first we bound the dimensions of the representations by $\frac{\dim(V_{\alpha_k}^n \otimes V_{\beta_k}^n \otimes V_{\delta_k}^n)}{\dim(V_{\epsilon_k}^n)} \leq \dim(V_{\alpha_k}^n \otimes V_{\beta_k}^n \otimes V_{\delta_k}^n) \leq (k+1)^{3 \binom{n}{2}}$. Then we upper bound the average of $\varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z)$ with the maximum of the absolute value. Together this gives

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \varphi_{\alpha_k}(a) \varphi_{\beta_k}(b) \varphi_{\delta_k}(d) \leq (k+1)^{3 \binom{n}{2}} \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z) \right|. \quad (287)$$

As this bound holds for all eigenvalues (a, b, d) , we can freely choose, for each k , a particular triple of eigenvalues (a^*, b^*, d^*) which ensures

$$\frac{k!}{\text{Tr}[e^*]^k} \varphi_{\alpha_k}(a^*) \varphi_{\beta_k}(b^*) \varphi_{\delta_k}(d^*) \geq \frac{1}{(k+1)^{3n}}. \quad (288)$$

where we are denoting the sum $\text{Tr}[a^*] + \text{Tr}[b^*] + \text{Tr}[d^*]$ with $\text{Tr}[e^*]$ by a slight abuse of notation as no eigenvalues e^* have been chosen here. That such a choice can always be made follows from the normalization condition

$$\sum_{\substack{\alpha, \beta, \delta \\ |\alpha| + |\beta| + |\delta| = k}} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) = \frac{(\text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d])^k}{k!}, \quad (289)$$

and noting that there are at most $(k+1)^{3n}$ such triples (α, β, δ) of Young diagrams in the sum (indeed it suffices to choose (a^*, b^*, d^*) to maximize $\varphi_{\alpha_k}(a) \varphi_{\beta_k}(b) \varphi_{\delta_k}(d)$). After this choice is made, we obtain the upper bound

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq (k+1)^{3 \binom{n}{2} + 3n} \frac{\text{Tr}[e^*]^k}{k!} \max_{(X,Y,Z) \in \mathcal{O}_{a^*} \times \mathcal{O}_{b^*} \times \mathcal{O}_{d^*}} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z) \right|. \quad (290)$$

Applying Eq. (274) from Lemma 60 here produces

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq (k+1)^{3\binom{n}{2}+3n+n(n-1)} \exp \left(-\frac{k}{4} \min_{(X,Y,Z) \in \mathcal{O}_{a^*} \times \mathcal{O}_{b^*} \times \mathcal{O}_{d^*}} R_{a^*b^*c^*d^*e^*f^*}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k} \right). \quad (291)$$

where $c^* = \exp(X+Y)$, $f^* = \exp(Y+Z)$, and $e^* = \exp(X+Y+Z)$.

Finally, since it was assumed that $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ are converging to non-tetrahedral eigenvalues, there must exist a sufficiently large K such that for all $k \geq K$ and all tetrahedral eigenvalues (a', b', c', d', e', f') , we have a separation $R_{a'b'c'd'e'f'}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k} \geq r$ for some positive constant $r > 0$. Since $(a^*, b^*, c^*, d^*, e^*, f^*)$ are tetrahedral eigenvalues, we can conclude for such sufficiently large $k \geq K$, $R_{a^*b^*c^*d^*e^*f^*}^{\alpha_k\beta_k\gamma_k\delta_k\epsilon_k\phi_k} \geq r$ and therefore

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq (k+1)^{3\binom{n}{2}+3n+n(n-1)} \exp \left(-\frac{k}{4} r \right), \quad (292)$$

which proves the desired exponential decay. $\square$

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